

## PRACTICE SHEET

1. What does the solution of the differential equation  $xy - ydx = 0$  represent?  
(a) Rectangular hyperbola  
(b) Straight line passing through (0, 0)  
(c) Parabola with vertex at (0, 0)  
(d) Circle with centre at (0, 0)
2. Which one of the following differential equations represents the system of circles touching  $y$  - axis at the origin?  
(a)  $\frac{dy}{dx} - x^2 - y^2$  (b)  $2xy \frac{dy}{dx} = y^2 - x^2$   
(c)  $2xy \frac{dy}{dx} = x^2 - y^2$  (d)  $\frac{dy}{dx} = y^2 - x^2$
3. What is the solution of the differential equation  $\frac{dy}{dx} = \frac{y}{(x+2y^3)}$ ?  
(a)  $y(1-xy) = cx$  (b)  $y^3 - x = cy$   
(c)  $x(1-xy) = cy$  (d)  $x(1+xy) = cy$
4. If  $y^2 = p(x)$  is a polynomial of degree 3, then what is  $2 \frac{d}{dx} \left[ y^3 \frac{d^2 y}{dx^2} \right]$  equal to?  
(a)  $p'(x)p''(x)$  (b)  $p''(x)p'''(x)$   
(c)  $p(x)p'''(x)$  (d) A constant
5. What is the degree of the equation  $\left[ \frac{d^2 y}{dx^2} \right] = \left[ y + \left( \frac{dy}{dx} \right)^2 \right]^{\frac{1}{4}}$   
(a) 1 (b) 2  
(c) 3 (d) 4
6. What are the order and degree respectively of the differential equation  $y = x \frac{dy}{dx} + \frac{dx}{dy}$ ?  
(a) 1, 1 (b) 1, 2  
(c) 2, 1 (d) 2, 2
7. What is the equation of the curve passing through the origin and satisfying the differential equation  $dy = (y \tan x + \sec x) dx$ ?  
(a)  $y = x \cos x$  (b)  $y \cos x = x$   
(c)  $xy = \cos x$  (d)  $y \sin x = x$
8. What is the solution of the differential equation  $\frac{dy}{dx} = \sec(x+y)$ ?  
(a)  $y + \tan(x+y) = c$  (b)  $y - \tan \left\{ \frac{x+y}{2} \right\} = c$   
(c)  $y + \tan \left\{ \frac{(x+y)}{2} \right\} = c$  (d)  $y + \tan \left\{ \frac{(x-y)}{2} \right\} = c$
9. For what value of  $k$  does the differential equation  $\frac{dy}{dx} = ky$  represent the law of natural decay?  
(a) -5 (b) 0  
(c) 0.01 (d) (10)-1
10. What is the solution of the differential equation  $(x+y)(dx-dy) = dx+dy$ ?  
(a)  $x+y + \ln(x+y) = c$   
(b)  $x-y + \ln(x+y) = c$   
(c)  $y-x + \ln(x+y) = c$   
(d)  $y-x - \ln(x-y) = c$
11. What is the degree of the differential equation,  $k \frac{d^2 y}{dx^2} = \left[ 1 + \left( \frac{dy}{dx} \right)^3 \right]^{3/2}$ , where  $k$  is a constant?  
(a) 1 (b) 2  
(c) 3 (d) 4
12. Under which one of the following conditions does the solution of  $\frac{dy}{dx} = \frac{ax+b}{cy+d}$  represent a parabola?  
(a)  $a=0, c=0$   
(b)  $a=1, b=2, c \neq 0$   
(c)  $a=0, c \neq 0, b \neq 0$   
(d)  $a=1, c=1$
13. A radioactive element disintegrates at a rate proportional to the quantity of substance  $Q$  present at any time  $t$ . What is the differential equation of the disintegration?  
(a)  $\frac{dQ}{dt} = -Q$  (b)  $\frac{dQ}{dt} = -kQ, k < 0$   
(c)  $\frac{dQ}{dt} = -kQ, k > 0$  (d)  $\frac{dQ}{dt} = Q$
14. What is the solution of the differential equation  $(x+y)(dx-dy) = dx+dy$ ?  
(a)  $2 \log(x+y) = c(y-x)$   
(b)  $(y-x) + \log(x+y) = c$   
(c)  $\left( \frac{y}{x} \right) + \left[ \log \left( \frac{y}{x} \right) \right] = c$   
(d) None of these
15. What is the only solution of the initial value problem  $y' = t(1+y)$ ,  $y(0) = 0$ ?  
(a)  $y = -1 + e^{t^2/2}$  (b)  $y = 1 + e^{t^2/2}$   
(c)  $y = -t$  (d)  $y = t$
16. What is the differential equation of the curve  $y = ax^2 + bx$ ?  
(a)  $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$   
(b)  $x^2 \frac{d^2 y}{dx^2} - y \left( \frac{dy}{dx} \right)^2 + 2 = 0$

- (c)  $(1-x^2)\frac{d^2y}{dx^2} - \left(y\frac{dy}{dx}\right)^2 = 0$   
 (d) None of the above
17. What is the degree of the differential equation  $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = k\frac{d^2y}{dx^2}$  ?  
 (a) 4 (b) 3  
 (c) 2 (d) 1
18. If  $f(x) = \sqrt{x + \sqrt{x + \sqrt{x + \sqrt{\dots\infty}}}}$ , then what is  $f'(x)$  equal  
 (a)  $\frac{1}{1-2f(x)}$  (b)  $\frac{1}{2f(x)-1}$   
 (c)  $\frac{1}{1+2f(x)}$  (d)  $\frac{1}{2+f(x)}$
19. What is the solution of the differential equation  $\frac{dy}{dx} = xy + x + y + 1$ ?  
 (a)  $y = \frac{x^2}{2} + x + c$   
 (b)  $\log(y+1) = \frac{x^2}{2} + x + c$   
 (c)  $y = x^2 + x + c$   
 (d)  $\log(y+1) = x^2 + x + c$
20. What are the order and degree, respectively of the differential equation  $\left(\frac{d^2y}{dx^2}\right)^{5/6} = \left(\frac{dy}{dx}\right)^{1/3}$  ?  
 (a) 2, 1 (b) 2, 5  
 (c)  $2, \frac{5}{6}$  (d)  $1, \frac{1}{3}$
21. Which one of the following is the undefined equation to family of circles having centre at the origin?  
 (a)  $(x^2 - y^2)\frac{dy}{dx} = 2xy$  (b)  $(x^2 + y^2)\frac{dy}{dx} = 2xy$   
 (c)  $\frac{dy}{dx}(x^2 + y^2)$  (d)  $x dx + y dy = 0$
22. What is the differential equation to family of parabolas having their vertices at the origin and foci on the X-axis?  
 (a)  $y=2x y'$  (b)  $x = 2yy'$   
 (c)  $xy = y'$  (d)  $x = yy'$
23. What is the general solution of  $(1+e^x)y dy = e^x dx$ ?  
 (a)  $y^2 = \ln[C^2(e^x + 1)^2]$  (d)  $dy = \ln[(e^x + 1)]$   
 (c)  $y^2 = \ln[C(e^x + 1)]$  (d) None of these
24. What is the degree of the differential equation  $\left(1 + \frac{dy}{dx}\right)^4 = \left(\frac{d^2y}{dx^2}\right)^2$   
 (a) 1 (b) 2  
 (c) 4 (d) 8
25. What does the equation  $y\frac{dy}{dx} + x = k$  (where, k is a constant) represents?  
 (a) A family of circles whose centre is on Y-axis  
 (b) A family of circles whose centre is on X-axis  
 (c) Touching the X-axis a family of circles  
 (d) None of the above
26. What is the degree of the differential equation  $\frac{d^2y}{dx^2} - \sqrt{1 + \left(\frac{dy}{dx}\right)^3} = 0$ ?  
 (a) 1 (b) 2  
 (c) 3 (d) 6
27. What is the solution of the differential equation  $3e^x \tan y dx + (1+e^x) \sec^2 y dy = 0$ ?  
 (a)  $(1+e^x) \tan y = C$  (b)  $(1+e^x)^3 \tan y = C$   
 (c)  $(1+e^x)^2 \tan y = C$  (d)  $(1+e^x) \sec^2 y = C$
28. What is the differential equation for  $y^2 = 4a(x-a)$ ?  
 (a)  $yy' - 2x yy' + y^2 = 0$   
 (b)  $yy'(yy' + 2x) + y^2 = 0$   
 (c)  $yy'(yy' - 2x) + y^2 = 0$   
 (d)  $yy' - 2x yy' + y = 0$
29. What is the solution of the differential equation  $a\left(x\frac{dy}{dx} + 2y\right) = xy\frac{dy}{dx}$ ?  
 (a)  $x^2 = kye^{\frac{y}{a}}$  (b)  $yx^2 = kye^{\frac{y}{a}}$   
 (c)  $y^2 x^2 = kye^{\frac{y}{a}}$  (d) None of these
30. The growth of a quantity  $N(t)$  at any constant  $t$  is given by  $\frac{dN(t)}{dt} = \alpha N(t)$ . Given that  $N(t) = ce^{kt}$ ,  $c$  is a constant. What is the value of  $\alpha$ ?  
 (a)  $c$  (b)  $k$   
 (c)  $c + k$  (d)  $c - k$

#### ANSWER KEYS

|     |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |
|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|
| 1.  | b | 2.  | b | 3.  | b | 4.  | c | 5.  | d | 6.  | b | 7.  | a | 8.  | b | 9.  | a | 10. | c |
| 11. | b | 12. | c | 13. | c | 14. | b | 15. | a | 16. | a | 17. | c | 18. | b | 19. | b | 20. | b |
| 21. | d | 22. | a | 23. | a | 24. | b | 25. | b | 26. | b | 27. | b | 28. | c | 29. | d | 30. | b |

## Solutions

### Sol. 1. (b)

Given that  $x \, dy - y \, dx = 0$   
Dividing both the sides by  $x^2$

$$\Rightarrow \frac{xy - y \, dx}{x^2} = 0$$

$$\Rightarrow d\left(\frac{y}{x}\right) = 0$$

$$\Rightarrow \frac{y}{x} = c \Rightarrow y = cx, \text{ where } c \text{ is a constant.}$$

Thus, it a straight line passing through (0,0).  
Alter :

Given,  $x \, dy - y \, dx = 0$

$$\Rightarrow \frac{dy}{y} = \frac{dx}{x}$$

Integration both the sides,  $\int \frac{dy}{y} = \int \frac{dx}{x} + \log c$

Log  $y = \log x + \log c$  where  $c$  is constant  
 $\Rightarrow y = cx$

### Sol. 2. (b)

Since, circle is touching  $y$  - axis at origin its center lies on  $x$  - axis. Let the centre be  $(a, 0)$ . Its radius =  $a$

$$(x - a)^2 + y^2 = a^2$$

$$x^2 + a^2 - 2ax + y^2 = a^2$$

$$\Rightarrow x^2 + y^2 - 2ax = 0$$

$$a = \frac{x^2 + y^2}{2x}$$

differentiating both side

$$\text{Now, } 2x + 2y \frac{dy}{dx} - 2a = 0$$

$$\Rightarrow 2x + 2y \frac{dy}{dx} - \frac{x^2 + y^2}{x} = 0$$

$$\Rightarrow 2x^2 + 2xy \frac{dy}{dx} - x^2 - y^2 = 0$$

$$\Rightarrow 2xy \frac{dy}{dx} = y^2 - x^2$$

### Sol. 3. (b)

$$y^3 - x = cy$$

$$\Rightarrow 3y^2 \frac{dy}{dx} - 1 = c \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} (3y^2 - c) = 1$$

$$\Rightarrow \frac{dy}{dx} \left( 3y^2 - \frac{y^3 - x}{y} \right) = 1$$

$$\Rightarrow \frac{dy}{dx} \left( \frac{3y^3 - y^3 + x}{y} \right) = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x + 2y^3}$$

### Sol. 4. (c)

Given that  $y^2 = p(x)$

Differentiating

$$\Rightarrow 2y y_1 = p'(x)$$

$$\Rightarrow 2y_1 = \frac{p'(x)}{y} \left[ \text{here } y_1 = \frac{dy}{dx} \right]$$

Differentiating again

$$\Rightarrow 2y_2 = \frac{y p''(x) - p'(x) y_1}{y^2}, \left[ y_2 = \frac{d^2 y}{dx^2} \right]$$

$$\Rightarrow 2y_2 = \frac{y p''(x) - \frac{p'(x) p'(x)}{2y}}{y^2}$$

$$= \frac{2y^2 p''(x) - p'(x)^2}{y^2}$$

$$\Rightarrow 2y^3 y_2 = 1/2 [2y^2 p''(x) - (p'(x))^2]$$

$$\Rightarrow 2 \frac{d}{dx} (y^3 y_2)$$

$$= 1/2 [2p'(x)p''(x) + 2p(x)p'''(x) - 2p'(x)p''(x)]$$

$$= p(x)p'''(x)$$

### Sol. 5. (d)

The given differential equation is:

This can be re - written as by squaring both the

sides to the power 4 to make it a polynomial of

derivative.

$$\left[ \frac{d^2 y}{dx^2} \right]^4 = \left[ y + \left( \frac{dy}{dx} \right)^2 \right]$$

Power of highest derivatives 4,

So, degree of the equation is 4.

### Sol. 6. (b)

The given differential equation is

$$y = x \frac{dy}{dx} + \frac{dx}{dy}$$

multiplying both the sides by  $\frac{dy}{dx}$

$$\text{We get } \left( \frac{dy}{dx} \right) y = x \left( \frac{dy}{dx} \right)^2 + 1$$

### Sol. 7. (a)

The differential equation

$dy = (y \tan x + \sec x) dx$  can be written as

$$\frac{dy}{dx} = y \tan x + \sec x$$

$$\text{or, } \frac{dy}{dx} - y \tan x = \sec x$$

Which is the form  $\frac{dy}{dx} + p(x) \cdot y = Q(x)$

Here  $P(x) = -\tan x$  and  $Q(x) = \sec x$

Integrating factor  $IF = e^{\int p(x) dx}$

$$IF = e^{\int -\tan x dx} = e^{-\int \frac{\sin x}{\cos x} dx}$$

$$\text{Putting } \cos x = t$$

$$-\sin x \, dx = dt$$

$$IF = e^{\int \frac{dt}{t}} = e^{\log_e t} = t = \cos x$$

The solution is

$$y \cdot Q(x) = \int I.F. Q(x) dx + c$$

$$\text{or, } y \cdot \sec x = \int \cos x \cdot \sec x \, dx + c$$

$$\text{or, } y \cdot \sec x = \int dx + c$$

$$\text{or, } y \cdot \sec x = x + c$$

Since the curve passes through the origin.

$$0 = 0 + c \Rightarrow c = 0$$

$$\text{and } y \sec x = x$$

or,  $y = x \cos x$

### Sol. 8. (b)

In the equation  $\frac{dy}{dx} = \sec(x+y)$

Let  $x + y = v$

$$\text{So, } 1 + \frac{dy}{dx} = \frac{dv}{dx} \text{ or } \frac{dy}{dx} = \frac{dv}{dx} - 1$$

$$\text{and } \frac{dv}{dx} - 1 = \sec v$$

$$\frac{dv}{dx} = 1 + \sec v = \frac{1 + \cos v}{\cos v}$$

$$\text{or, } \frac{\cos v}{1 + \cos v} dv = dx$$

$$\text{so, } \left( \frac{2 \cos^2 \frac{v}{2} \cdot 2 \sin^2 \frac{v}{2}}{2 \cos^2 \frac{v}{2}} \right) dv = dx$$

$$\left( 1 - \tan^2 \frac{v}{2} \right) dv = 2 dx$$

Integrating on both the sides

$$2 \int dv - \int \sec^2 v/2 \, dv = 2 \int dx + c_1$$

where  $c_1$  is a constant

$$2v - 2 \tan v/2 = 2x + c_1$$

$$\text{or, } 2x + 2y - 2 \tan \frac{x+y}{2} = 2x + c_1$$

$$y - \tan \left( \frac{x+y}{2} \right) = \frac{c_1}{2} = c, [c \text{ is constant}]$$

$$\text{so, } y - \tan \frac{x+y}{2} = c$$

### Sol. 9. (a)

$$\frac{dy}{dx} = ky \text{ or } \frac{dy}{y} = k dx$$

Integration both the sides

$$\int \frac{dy}{y} = k \int dx + \log c$$

$$\log y = kx + \log c \Rightarrow \log y - \log c = kx$$

$$\text{or } \log \left( \frac{y}{c} \right) = kx$$

$$\text{or, } \frac{y}{c} = e^{kx}$$

$$\text{or, } y = c \cdot e^{kx}$$

The equation will show a decay. If value of  $k$  is negative. Only option (a) shows negative value of  $k$ .

### Sol. 10. (c)

Differential equation is

$$(x+y)(dx-dy) = dx+dy$$

Dividing by  $dx$  on both the sides

$$(x+y) \left( 1 - \frac{dy}{dx} \right) = 1 + \frac{dy}{dx}$$

Putting  $x + y = v$

$$1 + \frac{dy}{dx} = \frac{dv}{dx} \text{ and } \frac{dy}{dx} = \frac{dv}{dx} - 1$$

The equation changes to

$$v \left\{ 1 - \left( \frac{dv}{dx} - 1 \right) \right\} = \frac{dv}{dx}$$

$$v \left( 2 - \frac{dv}{dx} \right) = \frac{dv}{dx}$$

$$2v - v \frac{dv}{dx} = \frac{dv}{dx}$$

$$2v = (1+v) \frac{dv}{dx}$$

$$\left( \frac{1+v}{v} \right) dv = 2dx$$

$$\text{or, } \left( \frac{1}{v} + 1 \right) dv = 2dx$$

integrating on both the sides

$$\int \frac{dv}{v} + \int dv = 2 \int dx + c$$

$$\text{Log } v + v = 2x + c$$

$$\text{Putting } v = x + y$$

$$\text{Log } (x + y) + x + y = 2x + c$$

$$\text{or, } \log (x + y) + y - x = c$$

$$\text{or, } y - x + \log (x + y) = c$$

**Sol. 11. (b)**

Squaring both the sides.

Here degree of differential equation is 2.

**Sol. 12. (c)**

$$\text{Given : } \frac{dy}{dx} = \frac{ax+b}{cy+d}$$

$$\text{Or, } (cy + d) dy = (ax + b) dx$$

Integrating both the sides.

$$c \int y dy + d \int dy = a \int x dx + b \int dx + K$$

[K is constant integration]

$$\text{or, c. } \frac{y^2}{2} + d.y = a \frac{x^2}{2} + b.x + K$$

$$\text{or, } cy^2 + 2d.y = ax^2 + 2b.x + 2K$$

This equation will represent a parabola when either, the coefficient of  $x^2$  or the coefficient of  $y^2$  is zero, but not both.

From the choice given,  $a=0$ ,  $c \neq 0$  and  $b \neq 0$ .

**Sol. 13. (c)**

A radioactive element disintegrates at a rate proportional to the quantity of substance Q present at any time t.

$$\Rightarrow \frac{dQ}{dt} \propto -Q$$

$$\Rightarrow \frac{dQ}{dt} = -KQ, k > 0 \text{ is a constant}$$

This is required differential equation

**Sol. 14. (b)**

Given differential equation is

$$(x + y) (dx - dy) = dx + dy$$

$$\Rightarrow (x + y) dx - (x + y) dy = dx + dy$$

$$\Rightarrow (x + y - 1) dx = (x + y + 1) dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{x + y - 1}{x + y + 1}$$

$$\text{Let } x + y = v \text{ and } \frac{dy}{dx} = \frac{dv}{dx} - 1$$

$$\therefore \frac{dv}{dx} - 1 = \frac{v-1}{v+1}$$

$$\Rightarrow \frac{dv}{dx} = \frac{v-1}{v+1} + 1$$

$$\Rightarrow \frac{dv}{dx} = \frac{v-1+v+1}{v+1}$$

$$\Rightarrow \frac{v+1}{2v} dv = dx$$

$$\Rightarrow \frac{1}{2} \int \frac{1}{v} dv + \frac{1}{2} \int \frac{1}{v} dv = \int 1 dx$$

$$\Rightarrow \frac{1}{2} v + \frac{1}{2} \log v = x + c_1$$

$$\Rightarrow x + y + \log (x + y) = 2x + c$$

$$(\because 2c_1 = c = \text{constant})$$

$$\Rightarrow (y - x) + \log (x + y) = c$$

**Sol. 15. (a)**

Given, equation is  $y' = t(1 + y)$

$$\text{i.e., } \frac{dy}{dt} = t(1 + y)$$

$$\Rightarrow \int \frac{1}{1+y} dy = \int t dt \Rightarrow \log(1+y) = \frac{t^2}{2} + c$$

As per initial conditions

$$y(0) = 0 \text{ when } t = 0, y = 0$$

$$\Rightarrow \log 1 = c \Rightarrow c = 0$$

$$\therefore \log(1+y) = \frac{t^2}{2} \Rightarrow 1+y = e^{t^2/2}$$

$$\Rightarrow y = -1 + e^{t^2/2}$$

Which is required solution.

**Sol. 16. (a)**

Given equation is :  $y = ax^2 + bx$

Differentiating w.r.t.x.

$$\Rightarrow \frac{dy}{dx} = 2ax + b$$

Differentiation g w.r.t.x,

$$\Rightarrow \frac{d^2y}{dx^2} = 2a$$

$$\Rightarrow a = \frac{1}{2} \frac{d^2y}{dx^2}$$

From (2) and (3)

$$b = \frac{dy}{dx} - x \cdot \frac{d^2y}{dx^2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = 2a$$

$$\Rightarrow a = \frac{1}{2} \frac{d^2y}{dx^2}$$

From (2) and (3)

$$\Rightarrow a = \frac{dy}{dx} - x \cdot \frac{d^2y}{dx^2}$$

Putting values of a and b in equation (1):

$$y = \frac{1}{2} \frac{d^2y}{dx^2} \cdot x^2 + x \left( \frac{dy}{dx} - x \frac{d^2y}{dx^2} \right)$$

$$\Rightarrow 2y = x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - 2x^2 \frac{d^2y}{dx^2}$$

$$\Rightarrow x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$$

**Sol. 17. (c)**

We square both side

Highest derivative has power = 2

Degree of differential equation = 2

**Sol. 18. (b)**

Given function is:

$$f(x) = \sqrt{x + \sqrt{x + \sqrt{x + \sqrt{\infty}}}}$$

$$f(x) = \sqrt{x + f(x)} \Rightarrow (f(x))^2 = x + f(x)$$

On differentiating both sides wrt. X, we get

$$2f(x) f'(x) = 1 + f'(x)$$

$$f'(x) \{2f(x) - 1\} = 1$$

$$f'(x) = \frac{1}{2f(x) - 1}$$

**Sol. 19. (b)**

The given differential equation is:

$$\frac{dy}{dx} = xy + x + y + 1 \Rightarrow \frac{dy}{dx} = (x+1)(y+1)$$

Separating variables,

$$\Rightarrow \frac{1}{(1+y)} dy = (x+1) dx \Rightarrow \log(1+y) = \frac{x^2}{2} + x + c$$

**Sol. 20. (b)**

Highest derivative has power of 5. So, the order and degree of given differential equation are 2 and 5 respectively.

**Sol. 21. (d)**

$$x^2 + y^2 = c^2$$

diff. w.r.t. x

$$2x + 2yy' = 0$$

$$x dx + y dy = 0$$

**Sol. 22. (a)**

$$y^2 = 4ax$$

$$2yy' = 4a$$

put this in equation (i)

$$y^2 = 2xy'x$$

$$y = 2xy'$$

**Sol. 23. (a)**

$$(1 + e^x) y dy = e^x dx$$

$$\int y dy = \int \frac{e^x}{1+e^x} dx$$

$$\frac{y^2}{2} = \log(1+e^x) + \log c$$

$$y^2 = 2 \log(1+e^x) + 2 \log c$$

$$y^2 = \log(1+e^x)^2 + \log c^2$$

$$y^2 = \log [c^2 (1+e^x)^2]$$

**Sol. 24. (b)**

**Sol. 25. (b)**

$$y \frac{dy}{dx} = -x + k$$

$$y dy = (-x + k) dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + kx$$

$$x^2 + y^2 - kx = 0$$

this equation represents a set of circles whose centre is on X axis.

**Sol. 26. (b)**

**Sol. 27. (b)**

$$3e^x \tan y dx + (1+e^x) \sec^2 y dy = 0$$

$$3e^x \tan y dx = - (1+e^x) \sec^2 y dy$$

$$\int \frac{3e^x}{(1+e^x)} dx = - \int \frac{\sec^2 y}{\tan y} dy$$

$$3 \log(1+e^x) = -\log \tan y + \log c$$

$$\tan y (1-e^x)^3 = c$$

**Sol. 28. (c)**

$$y^2 = 4a(x-a) \dots \dots (i)$$

diff. w.r.t. x

$$2yy' = 4a$$

$$a = \frac{yy'}{2}$$

on putting the value of a in eq(i) we get  
 $yy'(yy'-2x) + y^2 = 0$

**Sol. 29. (d)**

$$a \left( x \frac{dy}{dx} + 2y \right) = xy \frac{dy}{dx}$$

$$x(y-a) \frac{dy}{dx} = 2ay$$

$$\frac{(y-a)}{y} dy = \frac{2a}{x} dx$$

$$\int \left( 1 - \frac{a}{y} \right) dy = \int \frac{2a}{x} dx$$

$$y - a \log y = 2a \log x + a \log c$$

$$y = a \log y + 2a \log x + a \log c$$

$$y = a \log (x^2 y c)$$

$$x^2 y = k e^{y/a}$$

**Sol. 30. (b)**

$$\text{Given } N(t) = ce^{kt}$$

diff. both side w.r.t. t

$$\frac{dN(t)}{dt} = \frac{d}{dt} ce^{kt} = k(ce^{kt}) = k[N(t)]$$

but given

$$\frac{dN(t)}{dt} = \alpha N(t)$$

$$\text{so } \alpha = k$$

## NDA PYQ

1. What is the differential equation of all parabolas whose axes are parallel to Y-axis?  
 (a)  $\frac{d^3y}{dx^3} = 0$  (b)  $\frac{d^2x}{dy^2} = C$   
 (c)  $\frac{d^3x}{dy^3} = 1$  (d)  $\frac{d^3y}{dx^3} = C$   
**[NDA (I) - 2011]**
2. If the solution of the differential equation  $\frac{dy}{dx} = \frac{ax+3}{2y+f}$  represents a circle, then what is the value of a?  
 (a) 2 (b) 1  
 (c) -2 (d) -1  
**[NDA (I) - 2011]**
3. What is the degree of the differential equation  $\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} + 4 - 3\frac{d^2y}{dx^2} + 5\frac{dy}{dx} = 0$ ?  
 (a) 1 (b) 2  
 (c) 3 (d) 4  
**[NDA (I) - 2011]**
4. What does the differential equation  $y\frac{dy}{dx} + x = a$  (where a is a constant) represent?  
 (a) A set of circles having centre on Y axis  
 (b) A set of circles having centre on X axis  
 (c) A set of ellipse  
 (d) A pair of straight lines  
**[NDA (I) - 2011]**
5. What is the solution of the differential equation  $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$ ?  
 (a)  $\sin^{-1}y + \sin^{-1}x = c$   
 (b)  $\sin^{-1}y - \sin^{-1}x = c$   
 (c)  $2\sin^{-1}y + \sin^{-1}x = c$   
 (d)  $2\sin^{-1}y - \sin^{-1}x = c$   
 Where c is a constant  
**[NDA-2011(1)]**
6. What is the equation of curve passing through the point  $\left(0, \frac{\pi}{3}\right)$  satisfying the differential equation  $\sin x \cos y dx + \cos x \sin y dy = 0$ ?  
 (a)  $\cos x \cos y = \frac{\sqrt{3}}{2}$  (b)  $\sin x \sin y = \frac{\sqrt{3}}{2}$   
 (c)  $\sin x \sin y = \frac{1}{2}$  (d)  $\cos x \cos y = \frac{1}{2}$   
**NDA (II) - 2011]**
7. What is the degree of the differential equation  $\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} + 4 - 3\left(\frac{d^2y}{dx^2}\right) + 5\left(\frac{dy}{dx}\right) = 0$ ?  
 (a) 3 (b) 2  
 (c)  $\frac{2}{3}$  (d) not defined  
**[NDA-2011(2)]**
8. What is the degree of the differential equation  $y = x \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^{-1}$ ?  
 (a) 1  
 (b) 2  
 (c) -1  
 (d) Degree does not exist  
**[NDA (I) - 2012]**
9. Which one of the differential equation is not linear  
 (a)  $\frac{d^2y}{dx^2} + 4y = 0$  (b)  $x \frac{dy}{dx} + y = x^3$   
 (c)  $(x-y)^2 \frac{dy}{dx} = 9$  (d)  $\cos^2 x \frac{dy}{dx} + y = \tan x$   
**[NDA (I) - 2012]**
10. What is the solution of the differential equation  $\frac{dy}{dx} + \frac{y}{x} = 0$   
 (a)  $xy = c$  (b)  $x = cy$   
 (c)  $y = cx$  (d) None of these  
**[NDA (I) - 2012]**
11. What is the degree of the differential equation  $\frac{d^3y}{dx^3} + 2\left(\frac{d^2y}{dx^2}\right)^2 - \frac{dy}{dx} + y = 0$ .  
 (a) 6 (b) 3  
 (c) 2 (d) 1  
**[NDA (II) - 2012]**
12. Consider a differential equation of order m and degree n. Which one of the following pairs is not feasible?  
 (a) (3,2) (b) (2,3/2)  
 (c) (2,4) (d) (2,2)  
**[NDA (II) - 2012]**
13. What is the general solution of the differential equation  $e^x \tan y dx + (1-e^x) \sec^2 y dy = 0$ ?  
 (a)  $\sin y = C(1-e^x)$  (b)  $\cos y = C(1-e^x)$   
 (c)  $\cot y = C(1-e^x)$  (d) None of these  
**[NDA (II) - 2012]**
14. The differential equation representing the family of curves  $y = a \sin(\lambda x + \alpha)$  is:  
 (a)  $\frac{d^2y}{dx^2} + \lambda^2 y = 0$  (b)  $\frac{d^2y}{dx^2} - \lambda^2 y = 0$   
 (c)  $\frac{d^2y}{dx^2} + \lambda y = 0$  (d) None of these  
**[NDA (II) - 2012]**
15. For the differential equation  $\left(\frac{dy}{dx}\right)^2 - x \left(\frac{dy}{dx}\right) + y = 0$ , which one of the following is not its solution?  
 (a)  $y = x - 1$  (b)  $4y = x^2$   
 (c)  $y = x$  (d)  $y = -x - 1$   
**[NDA (II) - 2012]**
16. What is the general solution of the following differential equation  $x^2 dy + y^2 dx = 0$ ?  
 (a)  $x + y = c$  (b)  $xy = c$   
 (c)  $c(x+y) = y$  (d) None of these  
**[NDA (II) - 2012]**

17. The differential equation  $y \frac{dy}{dx} + x = a$  where  $a$  is a constant represents:  
 (a) A set of straight lines  
 (b) A set of ellipses  
 (c) A set of circles  
 (d) none of the above

[NDA (II) - 2012]

18. The degree and order respectively of the differential equation  $\frac{dy}{dx} = \frac{1}{x+y+1}$  are  
 (a) 1,1 (b) 1,2  
 (c) 2,1 (d) 2,2

[NDA (I) - 2013]

19. The general solution of the differential equation  $x \frac{dy}{dx} + y = 0$  is.  
 (a)  $xy = C$  (b)  $x = Cy$   
 (c)  $x + y = C$  (d)  $x^2 + y^2 = C$

[NDA (I) - 2013]

20. What is the degree of the differential equation  $\left(\frac{d^4y}{dx^4}\right)^{\frac{3}{5}} - 5\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 5 = 0$ ?  
 (a) 5 (b) 4  
 (c) 3 (d) 2

[NDA (I) - 2013]

21. The differential equation of the curve  $y = \sin x$  is:  
 (a)  $\frac{d^2y}{dx^2} + y \frac{dy}{dx} + x = 0$  (b)  $\frac{x^2y}{dx^2} + y = 0$   
 (c)  $\frac{d^2y}{dx^2} - y = 0$  (d)  $\frac{d^2y}{dx^2} + x = 0$

[NDA (I) - 2013]

22. The general solution of the differential equation  $\log\left(\frac{dy}{dx}\right) + x = 0$  is:  
 (a)  $y = e^{-x} + C$  (b)  $y = -e^{-x} + C$   
 (c)  $y = e^x + C$  (d)  $y = -e^x + C$

[NDA (I) - 2013]

23. What is the order of the differential equation  $\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} - \sin^2 y = 0$ ?  
 (a) 1 (b) 2  
 (c) 3 (d) Undefined

[NDA (II) - 2013]

24.  $y = 2\cos x + 3\sin x$  satisfies which of the following differential equations?

I.  $\frac{d^2y}{dx^2} + y = 0$

II.  $\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} = 0$

Select the correct answer using the codes given below:

- (a) Only I (b) Only II  
 (c) Both I and II (d) Neither I nor II

[NDA (II) - 2013]

25. What is the differential equation of all circles whose centres are at the origin is

- (a)  $\frac{dy}{dx} = \frac{y}{x}$  (b)  $\frac{dy}{dx} = \frac{x}{y}$   
 (c)  $\frac{dy}{dx} = -\frac{x}{y}$  (d) none of above

[NDA (II) - 2013]

26. What is the general solution of the differential equation  $x dy - y dx = y^2$ ?  
 (a)  $x = Cy$  (b)  $y^2 = Cx$   
 (c)  $x + xy - Cy = 0$  (d) None of these  
 (Where  $C$  is an arbitrary constant)

[NDA (I) - 2014]

**Directions (for next three):** The general solution of the differential equation  $(x^2 + x + 1) dy + (y^2 + y + 1) dx = 0$  is  $(x + y + 1) = A(1 + Bx + Cy + Dxy)$ , where  $B$ ,  $C$  and  $D$  are constants and  $A$  is parameter.

27. What is  $B$  equal to?  
 (a) -1 (b) 1  
 (c) 2 (d) None of these

[NDA (I) - 2014]

28. What is  $C$  equal to?  
 (a) 1 (b) -1  
 (c) 2 (d) None of these

[NDA (I) - 2014]

29. What is  $D$  equal to?  
 (a) -1 (b) 1  
 (c) -2 (d) None of these

[NDA (I) - 2014]

30. What is the solution of  $\frac{dy}{dx} + 2y = 1$  satisfying  $y(0) = 0$ ?  
 (a)  $y = \frac{1 - e^{-2x}}{2}$  (b)  $y = \frac{1 + e^{-2x}}{2}$   
 (c)  $y = 1 + e^x$  (d)  $y = \frac{1 + e^x}{2}$

[NDA (I) - 2014]

31. The solutions of  $\frac{dy}{dx} = |x|$  is  
 (a)  $y = \frac{x|x|}{2} + C$  (b)  $y = \frac{|x|}{2} + C$   
 (c)  $y = \frac{x^2}{2} + C$  (d)  $y = \frac{x^3}{2} + C$   
 (where,  $C$  is an arbitrary constant)

[NDA (I) - 2014]

32. What is the number of arbitrary constant in the particular solution of differential equation of third order?  
 (a) 0 (b) 1  
 (c) 2 (d) 3

[NDA (I) - 2014]

33. What is the equation of a curve passing through  $(0,1)$  and whose differential equation is given by  $dy = y \tan x dx$ ?  
 (a)  $y = \cos x$  (b)  $y = \sin x$   
 (c)  $y = \sec x$  (d)  $y = \operatorname{cosec} x$

[NDA (I) - 2014]

34. Consider the following statements in respect of the differential equation  $\frac{d^2y}{dx^2} + \cos\left(\frac{dy}{dx}\right) = 0$ .

I. The degree of the differential equation is not defined.

II. The order of the differential equation is 2.

Which of the above statement is/are correct?

- (a) Only I (b) Only II

- (c) Both I and II (d) Neither I nor II [NDA (I) - 2014]
35. What is the degree of the differential equation.  
 $\left(\frac{d^3y}{dx^3}\right)^{\frac{3}{2}} = \left(\frac{d^2y}{dx^2}\right)^2$  ?  
 (a) 1 (b) 2  
 (c) 3 (d) 4 [NDA (II) - 2014]
36. What is the solution of the equation  $\ln\left(\frac{dy}{dx}\right) + x = 0$   
 (a)  $y + e^x = C$  (b)  $y - e^{-x} = C$   
 (c)  $y + e^{-x} = C$  (d)  $y - e^x = C$  [NDA (II) - 2014]
37. Eliminating the arbitrary constants B and C in the expression  $y = \frac{2}{3C}(Cx - 1)^{\frac{3}{2}} + B$ , we get:  
 (a)  $x \left[1 + \left(\frac{dy}{dx}\right)^2\right] = \frac{d^2y}{dx^2}$   
 (b)  $2x \left(\frac{dy}{dx}\right) \frac{d^2y}{dx^2} = 1 + \left(\frac{dy}{dx}\right)^2$   
 (c)  $\left(\frac{dy}{dx}\right) \frac{d^2y}{dx^2} = 1$   
 (d)  $\left(\frac{dy}{dx}\right)^2 + 1 = \frac{d^2y}{dx^2}$  [NDA (II) - 2014]
38. Consider the following statements:  
 I. The general solution of  $\frac{dy}{dx} = f(x) + x$  is of the form  $y = g(x) + C$ , where C is an arbitrary constant,  
 II. The degree of  $\left(\frac{dy}{dx}\right)^2 = f(x)$  is 2.  
 Which of the above statements is/are correct?  
 (a) Only I (b) Only II  
 (c) Both I and II (d) Neither I nor II [NDA (I) - 2015]
39. What is the solution of the differential equation  $\frac{dx}{dy} + \frac{x}{y} - y^2 = 0$  ?  
 (a)  $xy = x^4 + C$  (b)  $xy = y^4 + C$   
 (c)  $4xy = y^4 + C$  (d)  $3xy = y^3 + C$   
 Where, C is an arbitrary constant. [NDA (I) - 2015]
40. What is the solution of the differential equation  $\frac{ydx - xdy}{y^2} = 0$  ?  
 (a)  $xy = C$  (b)  $y = cx$   
 (c)  $x + y = C$  (d)  $x - y = C$   
 Where, C is an arbitrary constant. [NDA (I) - 2015]
41. What is the solution of the differential equation  $\sin\left(\frac{dy}{dx}\right) - a = 0$ ?  
 (a)  $y = x \sin^{-1} a + C$  (b)  $x = y \sin^{-1} a + C$   
 (c)  $y = x + x \sin^{-1} a + C$  (d)  $y = \sin^{-1} a + C$   
 Where C is an arbitrary constant. [NDA (I) - 2015]
42. The degree of the differential equation  $\frac{dy}{dx} - x = \left(y - x \frac{dy}{dx}\right)^{-4}$  is:  
 (a) 2 (b) 3  
 (c) 4 (d) 5 [NDA (II) - 2015]
43. The solution of:  $\frac{dy}{dx} = \sqrt{1 - x^2 - y^2 + x^2 y^2}$  is  
 (a)  $\sin^{-1} y = \sin^{-1} x + c$   
 (b)  $2 \sin^{-1} y = \sqrt{1 - x^2} + \sin^{-1} x + c$   
 (c)  $2 \sin^{-1} y = x \sqrt{1 - x^2} + \sin^{-1} x + c$   
 (d)  $2 \sin^{-1} y = x \sqrt{1 - x^2} + \cos^{-1} x + c$   
 Where c is an arbitrary constant [NDA (II) - 2015]
44. The order and degree of the differential equation of parabolas having vertex at the origin and focus at (a,0) where  $a > 0$ , are respectively.  
 (a) 1,1 (b) 2,1  
 (c) 1,2 (d) 2,2 [NDA (II) - 2015]
45. The differential equation of the family of circles passing through the origin and having centres on the x-axis is:  
 (a)  $2xy \frac{dy}{dx} = x^2 - y^2$   
 (b)  $2xy \frac{dy}{dx} = y^2 - x^2$   
 (c)  $2xy \frac{dy}{dx} = x^2 + y^2$   
 (d)  $2xy \frac{dy}{dx} + x^2 + y^2 = 0$  [NDA (II) - 2015]
46. What are the order and degree respectively of the differential equation whose solution is  $y = cx + c^2 - 3c^{3/2} + 2$ , where c is a parameter?  
 (a) 1,2 (b) 2,2  
 (c) 1,3 (d) 1,4 [NDA (I) - 2016]
47. What are the degree and order respectively of the differential equation satisfying  $e^{\sqrt{1-x^2} + \sqrt{1-y^2}} = ce^x$ , (where  $c > 0, |x| < 1, |y| < 1$ )?  
 (a) 1,1 (b) 1,2  
 (c) 2,1 (d) 2,2 [NDA (II) - 2016]
48. What is the order of the differential equation  $\frac{dx}{dy} + \int y dx = x^3$ ?  
 (a) 1  
 (b) 2  
 (c) 3  
 (d) Cannot be determined [NDA (II) - 2016]
49. Which one of the following differential equations represents the family of straight lines which are at unit distance from the origin?  
 (a)  $\left(y - x \frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$



$$(b) \left( y + x \frac{dy}{dx} \right)^2 = 1 + \left( \frac{dy}{dx} \right)^2$$

$$(c) \left( y - x \frac{dy}{dx} \right)^2 = 1 + \left( \frac{dy}{dx} \right)^2$$

$$(d) \left( y + x \frac{dy}{dx} \right)^2 = 1 - \left( \frac{dy}{dx} \right)^2$$

[NDA (II) - 2016]

50. If  $x dy = y dx + y^2 dy$ ,  $y > 0$  and  $y(1)=1$ , then what is  $y(-3)$  equal to?  
 (a) 3 only (b) -1 only  
 (c) both -1 and 3 (d) neither -1 nor 3

[NDA-2016(2)]

51. What is the curve which passes through the point (1, 1) and whose slope is  $\frac{2y}{x}$ ?  
 (a) Circle (b) Parabola  
 (c) Ellipse (d) Hyperbola

[NDA-2016(2)]

52. What is the solution of the differential equation  $\ln \left( \frac{dy}{dx} \right) - a = 0$ ?  
 (a)  $y = xe^a + c$  (b)  $x = ye^a + c$   
 (c)  $y = \ln x + c$  (d)  $x = \ln y + c$

[NDA (I) - 2017]

53. What are the degree and order respectively for the differential equation  $y = x \left( \frac{dy}{dx} \right)^2 + \left( \frac{dx}{dy} \right)^2$ ?  
 (a) 1,2 (b) 2,1  
 (c) 1,4 (d) 4,1

[NDA (I) - 2017]

54. If  $x dy = y (dx + y dy)$ ;  $y(1) = 1$  and  $y(x) > 0$ , then what is  $y(-3)$  equal to?  
 (a) 3 (b) 2  
 (c) 1 (d) 0

[NDA (I) - 2017]

55. What is the differential equation corresponding to  $y^2 - 2ay + x^2 = a^2$  by eliminating  $a$ ?  
 (a)  $(x^2 - 2y^2) p^2 - 4pxy - x^2 = 0$   
 (b)  $(x^2 - 2y^2) p^2 + 4pxy - x^2 = 0$   
 (c)  $(x^2 + 2y^2) p^2 - 4pxy - x^2 = 0$   
 (d)  $(x^2 + 2y^2) p^2 - 4pxy + x^2 = 0$

$$\text{Where } p = \frac{dy}{dx}$$

[NDA (I) - 2017]

56. What is the general solution of differential equation  $yd x - (x + 2y^2) dy = 0$ ?  
 (a)  $x = y^2 + cy$  (b)  $x = 2cy^2$   
 (c)  $x = 2y^2 + cy$  (d) None of these

[NDA (I) - 2017]

57. The differential equation of minimum order by eliminating the arbitrary constants A and C in the equation  $y = A [\sin(x + C) + \cos(x + C)]$  is:  
 (a)  $y'' + (\sin x + \cos x) y' = 1$   
 (b)  $y'' = (\sin x + \cos x) y'$   
 (c)  $y'' = (y')^2 \sin x \cos x$   
 (d)  $y'' + y = 0$

[NDA (II) - 2017]

58. The order and degree of the differential equation:

$$\left[ 1 + \left( \frac{dy}{dx} \right)^2 \right]^3 = p^2 \left[ \frac{d^2 y}{dx^2} \right]^2 \text{ are respectively.}$$

- (a) 3 and 2 (b) 2 and 2  
 (c) 2 and 3 (d) 1 and 3

[NDA (II) - 2017]

59. The solution of the differential equation  $\frac{dy}{dx} = \frac{y\phi'(x) - y^2}{\phi(x)}$  is:

$$(a) y = \frac{x}{\phi(x) + c} \quad (b) y = \frac{\phi(x)}{x} + c$$

$$(c) y = \frac{\phi(x) + c}{x} \quad (d) y = \frac{\phi(x)}{x + c}$$

[NDA (II) - 2017]

60. The general solution of  $\frac{dy}{dx} = \frac{ax + h}{by + k}$  represents a circle only when:

- (a)  $a = b = 0$  (b)  $a = -b \neq 0$   
 (c)  $a = b \neq 0, h = k$  (d)  $a = b \neq 0$

[NDA (II) - 2017]

61. What are the order and degree, respectively, of the differential equation  $\left( \frac{d^3 y}{dx^3} \right)^2 = y^4 + \left( \frac{dy}{dx} \right)^5$ ?

- (a) 4,5 (b) 2,3  
 (c) 3,2 (d) 5,4

[NDA (I) - 2018]

62. What is the solution of  $(1+2x)dy - (1-2y)dx = 0$ ?

- (a)  $x - y - 2xy = c$  (b)  $y - x - 2xy = c$   
 (c)  $y + x - 2xt = c$  (d)  $x + y + 2xy = c$

[NDA (I) - 2018]

63. The order and degree of the differential equation  $y^2 = 4a(x-a)$ , where 'a' is an arbitrary constant, are respectively.

- (a) 1, 2 (b) 2, 1  
 (c) 2, 2 (d) 1, 1

[NDA (I) - 2018]

64. Which one of the following differential equations has a periodic solution?

- (a)  $\frac{d^2 x}{dt^2} + \mu x = 0$  (b)  $\frac{d^2 x}{dt^2} - \mu x = 0$   
 (c)  $x \frac{dx}{dt} + \mu t = 0$  (d)  $\frac{dx}{dt} + \mu xt = 0$

[NDA (I) - 2018]

65. What is the solution of the differential equation  $x dy - y dx = 0$ ?

- (a)  $xy = c$  (b)  $y = cx$   
 (c)  $x + y = c$  (d)  $x - y = c$

[NDA (I) - 2018]

66. What is the solution of the differential equation  $\frac{dx}{dy} = \frac{x+y+1}{x+y-1}$ ?

- (a)  $y - x + 4 \ln(x+y) = c$   
 (b)  $y + x + c \ln(x+y) = c$   
 (c)  $y - x + \ln(x+y) = c$   
 (d)  $y + x + 2 \ln(x+y) = c$

[NDA (II) - 2018]

67. If  $y = \sin(\ell nx)$ , then which one of the following is correct?

(a)  $\frac{d^2y}{dx^2} + y = 0$  (b)  $\frac{d^2y}{dx^2} = 0$   
 (c)  $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$  (d)  $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = 0$

[NDA (II) - 2018]

68. What is the solution of the differential equation  $\log\left(\frac{dy}{dx}\right) = ax + by$ ?

(a)  $a e^{ax} + b e^{by} = c$  (b)  $\frac{1}{a} e^{ax} + \frac{1}{b} e^{by} = c$   
 (c)  $a e^{ax} + b e^{-by} = c$  (d)  $\frac{1}{a} e^{ax} + \frac{1}{b} e^{by} = c$

[NDA (II) - 2018]

69. What is the order of the differential equation whose solution is  $y = a \cos x + b \sin x + ce^{-x} + d$ , where  $a, b, c$  and  $d$  are arbitrary constants?

(a) 1 (b) 2  
 (c) 3 (d) 4

[NDA (II) - 2018]

70. The equation of the curve passing through the point  $(-1, -2)$  which satisfies  $\frac{dy}{dx} = -x^2 - \frac{1}{x^3}$  is.

(a)  $17x^2y - 6x^2 + 3x^5 - 2 = 0$   
 (b)  $6x^2y + 17x^2 + 2x^5 - 3 = 0$   
 (c)  $6xy - 2x^2 + 17x^5 + 3 = 0$   
 (d)  $17x^2y + 6xy - 3x^5 + 5 = 0$

[NDA (II) - 2018]

71. The differential equation of the family of curves  $y = p \cos(ax) + q \sin(ax)$ , where  $p, q$  are arbitrary constants, is:

(a)  $\frac{d^2y}{dx^2} - a^2y = 0$  (b)  $\frac{d^2y}{dx^2} - ay = 0$   
 (c)  $\frac{d^2y}{dx^2} + ay = 0$  (d)  $\frac{d^2y}{dx^2} + a^2y = 0$

[NDA (II) - 2018]

72. What is the general solution of the differential equation  $\frac{dy}{dx} + \frac{x}{y} = 0$ ?

(a)  $x^2 + y^2 = c$  (b)  $x^2 - y^2 = c$   
 (c)  $x^2 + y^2 = cxy$  (d)  $x + y = c$

[NDA (I) - 2019]

73. Consider the following in respect of the differential equation  $\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 + 9y = x$

1. degree of the differential equation is 1  
 2. order of the differential equation is 2  
 Which of the above statement is/are correct?

(a) 1 only (b) 2 only  
 (c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2019]

74. The differential equation of the system of circles touching the  $y$  axis at origin is

(a)  $x^2 + y^2 - 2xy \frac{dy}{dx} = 0$  (b)  $x^2 + y^2 + 2xy \frac{dy}{dx} = 0$   
 (c)  $x^2 - y^2 + 2xy \frac{dy}{dx} = 0$  (d)  $x^2 - y^2 - 2xy \frac{dy}{dx} = 0$

[NDA (I) - 2019]

75. If  $y = a \cos 2x + b \sin 2x$ , then

(a)  $\frac{d^2y}{dx^2} + y = 0$  (b)  $\frac{d^2y}{dx^2} + 2y = 0$   
 (c)  $\frac{d^2y}{dx^2} - 4y = 0$  (d)  $\frac{d^2y}{dx^2} + 4y = 0$

[NDA (I) - 2019]

76. What is the solution of the differential equation  $\frac{dy}{dx} = \cos(y-x) + 1$ ?

(a)  $e^x [\sec(y-x) - \tan(y-x)] = c$   
 (b)  $e^x [\sec(y-x) + \tan(y-x)] = c$   
 (c)  $e^x \sec(y-x) \tan(y-x) = c$   
 (d)  $e^x = \sec(y-x) \tan(y-x)$

[NDA (I) - 2019]

77. What is the degree of differential equation

$\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^2 - x^2 \left(\frac{d^2y}{dx^2}\right) = 0$   
 (a) 1 (b) 2  
 (c) 3 (d) 4

[NDA (II) - 2019]

78. The differential equation which represents the family of curves given by  $\tan y = c(1 - e^x)$  is

(a)  $e^x \tan y dx + (1 - e^x) dy = 0$   
 (b)  $e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$   
 (c)  $e^x (1 - e^x) dx + \tan y dy = 0$   
 (d)  $e^x \tan y dy + (1 - e^x) dx = 0$

[NDA (II) - 2019]

79. Which one of the following is the differential equation that represents the family of curves  $y = \frac{1}{2x^2 - c}$  where  $c$  is an arbitrary constant?

(a)  $\frac{dy}{dx} = 4xy^2$  (b)  $\frac{dy}{dx} = \frac{1}{y}$   
 (c)  $\frac{dy}{dx} = x^2y$  (d)  $\frac{dy}{dx} = -4xy^2$

[NDA-2019(2)]

80. The function  $u(x, y) = c$  which satisfies the differential equation  $x(dx - dy) + y(dy - dx) = 0$ , is

(a)  $x^2 + y^2 = xy + c$  (b)  $x^2 + y^2 = 2xy + c$   
 (c)  $x^2 - y^2 = xy + c$  (d)  $x^2 - y^2 = 2xy + c$

[NDA - 2020]

81. What is the solution of the differential equation

$\ln\left(\frac{dy}{dx}\right) = x$ ?  
 (a)  $y = e^x + c$  (b)  $y = e^{-x} + c$   
 (c)  $y = \ln x + c$  (d)  $y = 2 \ln x + c$

[NDA - 2020]

82. The solution of the differential equation  $dy = (1 + y^2) dx$  is

(a)  $y = \tan x + c$   
 (b)  $y = \tan(x + c)$   
 (c)  $\tan^{-1}(y + c) = x$   
 (d)  $\tan^{-1}(y + c) = 2x$

[NDA - 2020]

83. The order and degree of the differential equation

$k \frac{dy}{dx} = \int \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{2}{3}}$  are respectively  
 (a) 1 and 1 (b) 2 and 3

- (c) 2 and 4 (d) 1 and 4 [NDA - 2020]
84. If the general solution of a differential equation is  $y^2 + 2cy - cx + c^2 = 0$ , where  $c$  is an arbitrary constant, then what is the order of the differential equation?  
(a) 1 (b) 2  
(c) 3 (d) 4
85. What is the degree of following differential equation?  
 $x = \sqrt{1 + \frac{d^2y}{dx^2}}$   
(a) 1 (b) 2  
(c) 3 (d) Degree is not defined [NDA (I) 2021]
86. Which of the following differential equations has the general solution  $y = ae^x + be^{-x}$ ?  
(a)  $\frac{d^2y}{dx^2} + y = 0$  (b)  $\frac{d^2y}{dx^2} - y = 0$   
(c)  $\frac{d^2y}{dx^2} + y = 1$  (d)  $\frac{dy}{dx} - y = 0$  [NDA (I) 2021]
87. The solution of the differential equation  $\ln\left(\frac{dy}{dx}\right) + y = x$  is  
(a)  $e^x + e^y = c$  (b)  $e^{x+y} = c$   
(c)  $e^x - e^y = c$  (d)  $e^{x-y} = c$  [NDA (I) 2021]
88. A particle starts from origin with a velocity (in m/sec) given by the equation  $\frac{dx}{dt} = x + 1$ . The time (in second) taken by the particle to transverse a distance of 24 m is  
(a)  $\ln 24$  (b)  $\ln 5$   
(c)  $2 \ln 5$  (d)  $2 \ln 4$  [NDA (I) 2021]
89. What is the order of the differential equation of all ellipses whose axes are along the coordinate axes?  
(a) 1 (b) 2  
(c) 3 (d) 4 [NDA (II) 2021]
90. What is the degree of the differential equation of all circles touching both the coordinate axes in the first quadrant?  
(a) 1 (b) 2  
(c) 3 (d) 4 [NDA (II) 2021]
91. What is the differential equation of  $y = A - \frac{B}{x}$ ?  
(a)  $xy_2 + y_1 = 0$  (b)  $xy_2 + 2y_1 = 0$   
(c)  $xy_2 - 2y_1 = 0$  (d)  $2xy_2 + y_1 = 0$  [NDA (II) 2021]
92. If  $\frac{dy}{dx} = (\ln 5)y$  with  $y(0) = \ln 5$ , then what is  $y(1)$  equal to?  
(a) 0 (b) 5  
(c)  $2 \ln 5$  (d)  $5 \ln 5$  [NDA (I) 2022]
93. What does the equation  $x \frac{dy}{dx} - 2y = 0$  represent?  
(a) A family of straight lines  
(b) A family of circles  
(c) A family of parabolas  
(d) A family of ellipses [NDA (I) 2022]

94. What is the degree of the differential equation  $1 + \left(\frac{dy}{dx}\right)^2 = \left(\frac{d^2y}{dx^2}\right)^{4/3}$ ?  
(a)  $\frac{4}{3}$  (b) 2  
(c) 3 (d) 4 [NDA (I) 2022]
95. A radioactive substance decays at a rate proportional to the amount of substance present. If half of the substance decays in 100 years, then what is the decay constant (proportionality constant)?  
(a)  $\frac{\ln 2}{100}$  (b)  $\frac{\ln 5}{100}$   
(c)  $\frac{\ln 10}{100}$  (d)  $\frac{2 \ln 2}{100}$  [NDA (I) 2022]
96. Consider the following statements:  
1. The degree of the differential equation  $\frac{dy}{dx} + \cos\left(\frac{dy}{dx}\right) = 0$  is 1.  
2. The order of the differential equation  $\left(\frac{d^2y}{dx^2}\right)^3 + \cos\left(\frac{dy}{dx}\right) = 0$  is 2.  
Which of the statements given above is/are correct?  
(a) 1 only (b) 2 only  
(c) Both 1 and 2 (d) neither 1 nor 2 [NDA 2022 (II)]
97. What is the differential equation of the family of parabolas having vertex at origin and axis along positive y-axis?  
(a)  $x \frac{dy}{dx} + 2y = 0$  (b)  $x \frac{dy}{dx} - 2y = 0$   
(c)  $y \frac{dx}{dy} + 2x = 0$  (d)  $y \frac{dx}{dy} - 2x = 0$  [NDA 2022 (II)]
98. What is the solution of the differential equation  $(dy - dx) + \cos x (dy + dx) = 0$ ?  
(a)  $y = \tan\left(\frac{x}{2}\right) - x + c$  (b)  $y = \frac{1}{2} \tan\left(\frac{x}{2}\right) - x + c$   
(c)  $y = 2 \tan\left(\frac{x}{2}\right) - x + c$  (d)  $y = \tan\left(\frac{x}{2}\right) - 2x + c$  [NDA 2022 (II)]
99. If  $y = (x^x)^x$ , then which one of the following is correct?  
(a)  $\frac{dy}{dx} + xy(1 + 2 \ln x) = 0$  (b)  $\frac{dy}{dx} - xy(1 + 2 \ln x) = 0$   
(c)  $\frac{dy}{dx} - 2xy(1 + \ln x) = 0$  (d)  $\frac{dy}{dx} + 2xy(1 + \ln x) = 0$  [NDA 2022 (II)]
100. A solution of the differential equation  $\left(\frac{dy}{dx}\right)^2 - x \frac{dy}{dx} = 0$  is:  
(a)  $y = 2x$  (b)  $y = 2x + 4$   
(c)  $y = x^2 - 1$  (d)  $y = \frac{x^2 - 2}{2}$  [NDA - 2023 (1)]

Consider the following for the next two (02) items that follow:

Suppose E is the differential equation representing family of curves  $y^2 = 2cx + 2c\sqrt{c}$ , where c is a positive parameter.

101. What is the order of the differential equation?

- (a) 1 (b) 2  
(c) 3 (d) 4

[NDA – 2023 (1)]

102. What is the degree of the differential equation?

- (a) 2 (b) 3  
(c) 4 (d) degree does not exist

[NDA – 2023 (1)]

Consider the following for the next (02) items that follow

A parabola passes through (1, 2) and satisfies the differential equation  $\frac{dy}{dx} = \frac{2y}{x}$ ,  $x > 0, y > 0$

103. What is the directrix of the parabola?

- (a)  $y = -1/8$  (b)  $y = 1/8$   
(c)  $x = -1/8$  (d)  $x = 1/8$

[NDA-2023 (2)]

104. What is the length of latus rectum of the parabola?

- (a) 1 (b) 1/2  
(c) 1/4 (d) 1/8

[NDA-2023 (2)]

105. What are the order and the degree respectively of the

differential equation  $x^2 \left( \frac{d^3y}{dx^3} \right)^2 + \left( \frac{dy}{dx} \right)^4 + \sin x = 0$ ?

- (a) 3,4 (b) 1,4  
(c) 2,2 (d) 3,2

[NDA-2023 (2)]

106. What is the differential equation of all parabolas of the type  $y^2 = 4a(x-b)$ ?

(a)  $\frac{d^2y}{dx^2} + \left( \frac{dy}{dx} \right)^2 = 0$  (b)  $\frac{d^2y}{dx^2} + x^2 \left( \frac{dy}{dx} \right)^2 = 0$

(c)  $y^2 \frac{d^2y}{dx^2} + \left( \frac{dy}{dx} \right)^2 = 0$  (d)  $y \frac{d^2y}{dx^2} + \left( \frac{dy}{dx} \right)^2 = 0$

[NDA-2023 (2)]

107. Let  $y_1(x)$  and  $y_2(x)$  be two solutions of the differential equation  $\frac{dy}{dx} = x$ . If  $y_1(0) = 0$  and

$y_2(0) = 4$ , then what is the number of points of intersection of the curves  $y_1(x)$  and  $y_2(x)$ ?

- (a) no point (b) one point  
(c) two points (d) More than two points

[NDA-2024 (1)]

108. The differential equation, representing the curve  $y = e^x (a \cos x + b \sin x)$  where a and b are arbitrary constants, is

(a)  $\frac{d^2y}{dx^2} + 2y = 0$  (b)  $\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + 2y = 0$

(c)  $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$  (d)  $\frac{d^2y}{dx^2} + y = 0$

[NDA-2024 (1)]

109. What are the order and degree respectively of the differential equation

$$\left\{ 2 - \left( \frac{dy}{dx} \right) \right\}^{0.6} = \frac{d^2y}{dx^2}$$

- (a) 2, 2  
(c) 5, 2

- (b) 2, 3  
(d) 2, 5

[NDA-2024 (1)]

110. If  $\frac{dy}{dx} = 2e^x y^3$ ,  $y(0) = \frac{1}{2}$  then what is  $4y^2(2 - e^x)$  equal to?

- (a) 1 (b) 2  
(c) 3 (d) 4

[NDA-2024 (1)]

Direction: Consider the following for the two items given below

Let  $ydx + (x - y^3)dy = 0$  be a differential equation.

111. What are the order and degree respectively of the differential equation?

- (a) 1 and 1 (b) 1 and 2  
(c) 2 and 1 (d) 1 and 3

[NDA-2024 (2)]

112. What is the solution of differential equation ?

- (a)  $y^4 + 2x = c$  (b)  $y^4 + 3x = c$   
(c)  $2xy^4 + x = c$  (d)  $4xy - y^4 = c$

[NDA-2024 (2)]

Consider the following for the one (01) items that follow:

The slope of the tangent to the curve  $y = f(x)$  at  $(x, f(x))$  is 4 for every real number x and the curve passes through the origin.

113. What is the nature of the curve?

- (a) A straight line passing through (1, 4)  
(b) A straight line passing through (-1, 4)  
(c) A parabola with vertex at origin and focus at (2, 0)  
(d) A parabola with vertex at origin and focus at (1, 0)

[NDA-2025 (1)]

114. What is the degree of the differential equation

$$\left( \frac{d^2y}{dx^2} \right)^{\frac{3}{2}} = \left( \frac{dy}{dx} \right)^{\frac{5}{2}}$$

- (a) 3 (b) 2  
(c) 5/2 (d) 3/2

[NDA-2025 (2)]

115. Consider the following statements

I.  $y = xe^{2x}$  is the solution of  $\frac{dy}{dx} = y \left( 2 + \frac{1}{x} \right)$

II.  $y = x \ln|x| + cx$  is the solution of  $\frac{dy}{dx} = \frac{x+y}{x}$

Which of the statements given above is/are correct?

- (a) I only (b) II only  
(c) Both I and II (d) Neither I nor II

[NDA-2025 (2)]

116. If k is an arbitrary constant. Then what is the general solution of the equation  $(x+y)^2 \frac{dy}{dx} = k^2$ ?

(a)  $y + x = \tan(x+c) + k$  (b)  $x + y = k \tan \left( \frac{y-c}{k} \right)$

(c)  $x - y = k \tan \left( \frac{y-c}{k} \right)$  (d)  $y - x = \tan(x+c) + k$

[NDA-2025 (2)]

| ANSWER KEY |   |      |   |      |   |      |   |      |   |      |   |      |   |      |   |
|------------|---|------|---|------|---|------|---|------|---|------|---|------|---|------|---|
| 1.         | a | 2.   | c | 3.   | b | 4.   | b | 5.   | a | 6.   | d | 7.   | b | 8.   | b |
| 11.        | d | 12.  | b | 13.  | d | 14.  | a | 15.  | c | 16.  | c | 17.  | c | 18.  | a |
| 21.        | b | 22.  | b | 23.  | a | 24.  | a | 25.  | c | 26.  | d | 27.  | a | 28.  | b |
| 31.        | a | 32.  | d | 33.  | c | 34.  | c | 35.  | c | 36.  | c | 37.  | b | 38.  | c |
| 41.        | a | 42.  | d | 43.  | c | 44.  | a | 45.  | b | 46.  | d | 47.  | a | 48.  | b |
| 51.        | d | 52.  | a | 53.  | d | 54.  | a | 55.  | a | 56.  | c | 57.  | d | 58.  | b |
| 61.        | c | 62.  | a | 63.  | a | 64.  | a | 65.  | b | 66.  | c | 67.  | c | 68.  | d |
| 71.        | d | 72.  | a | 73.  | c | 74.  | c | 75.  | d | 76.  | a | 77.  | c | 78.  | b |
| 81.        | a | 82.  | b | 83.  | b | 84.  | a | 85.  | a | 86.  | b | 87.  | c | 88.  | c |
| 91.        | b | 92.  | d | 93.  | c | 94.  | d | 95.  | a | 96.  | b | 97.  | b | 98.  | b |
| 101.       | a | 102. | b | 103. | a | 104. | b | 105. | d | 106. | d | 107. | a | 108. | c |
| 111.       | a | 112. | d | 113. | a | 114. | a | 115. | c | 116. | b |      |   |      |   |

## Solutions

### Sol.1. (a)

parabola whose axis is parallel to y axis

$$(x-h)^2 = 4a(y-k)$$

diff. w.r.t. x

$$2(x-h) = 4a \frac{dy}{dx}$$

again diff. w.r.t. x

$$2 = 4a \frac{d^2y}{dx^2}$$

$$\frac{d^3y}{dx^3} = 0$$

### Sol.2. (c)

$$(2y+f) dy = (ax+3) dx$$

integrate both sides

$$y^2 + fy = a \frac{x^2}{2} + 3x + c$$

for circle coefficient of  $x^2$  and  $y^2$  should be same  
so  $a = -2$

### Sol.3. (b)

### Sol.4. (b)

$$y \frac{dy}{dx} = -x + a$$

$$y dy = (-x+a) dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + ax$$

$$x^2 + y^2 - ax = 0$$

this equation represents a set of circles whose centre is on X axis.

### Sol.5. (a)

$$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}} = 0$$

$$\frac{dy}{\sqrt{1-y^2}} = -\frac{dx}{\sqrt{1-x^2}}$$

$$\sin^{-1} y = -\sin^{-1} x + c$$

### Sol.6. (d)

$$\sin x \cos y dx + \cos x \sin y dy = 0$$

$$-\tan y dy = \tan x dx$$

integrate both sides

$$\log \cos y = \log \sec x + \log c$$

$$\cos y = c \sec x$$

$$\cos x \cos y = c$$

put given point in equation then  $c = 1/2$

$$\cos x \cos y = 1/2$$

### Sol.7. (b)

### Sol.8. (b)

### Sol.9. (a)

because second order differential equation.

### Sol.10. (a)

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\int \frac{dy}{y} = -\int \frac{dx}{x}$$

$$\log y = -\log x + \log c$$

$$\log y + \log x = \log c$$

$$xy = c$$

### Sol.11. (d)

### Sol.12. (b)

degree cannot be fractional

### Sol.13. (d)

$$e^x \tan y dx + (1-e^x) \sec^2 y dy = 0$$

$$e^x \tan y dx = - (1-e^x) \sec^2 y dy$$

$$-\int \frac{e^x}{(1-e^x)} dx = \int \frac{\sec^2 y}{\tan y} dy$$

$$\log(1-e^x) = \log \tan y + \log c$$

$$(1-e^x) = \tan y$$

$$\tan y = C(1-e^x)$$

### Sol.14. (a)

$$y = a \sin(\lambda x + \alpha) \quad \dots(i)$$

$$\frac{dy}{dx} = a \lambda \cos(\lambda x + \alpha) \quad \dots(ii)$$

$$\frac{d^2y}{dx^2} = -a \lambda^2 \sin(\lambda x + \alpha) \quad \dots(iii)$$

$$(i) + (iii)$$

$$\frac{d^2y}{dx^2} + \lambda^2 y = 0$$

### Sol.15. (c)

This question can solve only by options and only c option satisfying the differential equation.

### Sol.16. (c)

$$x^2 dy + y^2 dx = 0$$

$$\int \frac{dy}{y^2} = -\int \frac{dx}{x^2}$$

$$-\frac{1}{y} = \frac{1}{x} + c$$

$$-c = \frac{1}{x} + \frac{1}{y} \Rightarrow -cxy = x + y$$

$$xy = C(x+y) \text{ put } -1/c = C]$$

### Sol.17. (c)

$$y \frac{dy}{dx} = -x + a$$

$$y dy = (-x+a) dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + ax$$

$$x^2 + y^2 - ax = 0$$

this equation represents a set of circles.

### Sol.18. (a)

order and degree both are 1

### Sol.19. (a)

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\int \frac{dy}{y} = -\int \frac{dx}{x}$$

$$\log y = -\log x + \log c$$

$$xy = c$$

### Sol.20. (b)

### Sol.21. (b)

$$y = \sin x \quad \dots(i)$$

$$\frac{dy}{dx} = \cos x \quad \dots(ii)$$

$$\frac{d^2y}{dx^2} = -\sin x \quad \dots(iii)$$

$$(i) + (iii)$$

$$\frac{d^2y}{dx^2} + y = 0$$

### Sol.22. (b)

$$\ln \left( \frac{dy}{dx} \right) + x = 0$$

$$\frac{dy}{dx} = e^{-x} \Rightarrow dy = e^{-x} dx$$

integrating both sides

$$y = -e^{-x} + c$$

$$y + e^{-x} = c$$

### Sol.23. (a)

### Sol.24. (a)

$$y = 2 \cos x + 3 \sin x \quad \dots(i)$$

$$\frac{dy}{dx} = -2 \sin x + 3 \cos x \quad \dots(ii)$$

$$\frac{d^2y}{dx^2} = -2\cos x - 3\sin x \quad \dots(iii)$$

(i) + (iii)

$$\frac{d^2y}{dx^2} + y = 0$$

only I statement is correct.

**Sol.25. (c)**

$$x^2 + y^2 = c^2$$

diff. w.r.t. x

$$2x + 2yy' = 0$$

$$x + yy' = 0$$

$$y' = -x/y$$

**Sol.26. (d)**

$$\frac{ydx - xdy}{y^2} = 1$$

$$d\left(\frac{x}{y}\right) = 1$$

integrate both sides

$$\int d\left(\frac{x}{y}\right) = x + c$$

$$\frac{x}{y} = x + c \Rightarrow x = xy + cy$$

**Sol.27. (a)**

$$(x^2 + x + 1)dy + (y^2 + y + 1)dx = 0$$

$$\int \frac{dx}{x^2 + x + 1} = -\int \frac{dy}{y^2 + y + 1}$$

$$\int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} + \int \frac{dy}{\left(y + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 0$$

$$\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2y+1}{\sqrt{3}}\right) = \frac{2}{\sqrt{3}} \tan^{-1} c$$

$$\tan^{-1}\left(\frac{\frac{2x+1}{\sqrt{3}} + \frac{2y+1}{\sqrt{3}}}{1 - \frac{2x+1}{\sqrt{3}} \cdot \frac{2y+1}{\sqrt{3}}}\right) = \tan^{-1} c$$

by solving it

$$x + y + 1 = \frac{c}{\sqrt{3}}(1 - x - y - 2xy)$$

$$(x + y + 1) = A(1 + Bx + Cy + Dxy)$$

$$B = -1$$

**Sol.28. (b)**

$$C = -1$$

**Sol.29. (c)**

$$D = -2$$

**Sol.30. (a)**

$$\frac{dy}{dx} + 2y = 1 \Rightarrow \frac{dy}{dx} = 1 - 2y$$

$$\int \frac{dy}{1-2y} = \int dx$$

$$-\frac{1}{2} \log |1-2y| = x + c$$

$$\text{put } x = y = 0$$

$$\text{then } c = 0$$

$$-\frac{1}{2} \log |1-2y| = x$$

$$y = \frac{1 - e^{-2x}}{2}$$

**Sol.31. (a)**

$$\int dy = \int |x| dx$$

$$y = \frac{x|x|}{2} + C$$

**Sol.32. (d)**

**Sol.33. (c)**

$$dy = y \tan x$$

$$\int \frac{dy}{y} = \int \tan x dx$$

$$\log y = \log(\sec x) + \log c$$

$$y = c \sec x$$

$$\text{put } x = 0 \text{ and } y = 1 \text{ then } c = 1$$

$$\text{equation is } y = \sec x$$

**Sol.34. (c)**

**Sol.35. (c)**

after squaring both sides degree is 3

**Sol.36. (c)**

$$\ln\left(\frac{dy}{dx}\right) + x = 0$$

$$\frac{dy}{dx} = e^{-x} \Rightarrow dy = e^{-x} dx$$

integrating both sides

$$y = -e^{-x} + c$$

$$y + e^{-x} = c$$

**Sol.37. (b)**

$$y = \frac{2}{3C}(Cx - 1)^{\frac{3}{2}} + B,$$

diff. w.r.t. x

$$\frac{dy}{dx} = (Cx - 1)^{\frac{1}{2}}$$

$$\left(\frac{dy}{dx}\right)^2 = Cx - 1$$

$$\left(\frac{dy}{dx}\right)^2 + 1 = Cx$$

$$2\left(\frac{dy}{dx}\right)\left(\frac{d^2y}{dx^2}\right) = C$$

from eq. (i)

$$\left(\frac{dy}{dx}\right)^2 + 1 = \left[2x\left(\frac{dy}{dx}\right)\left(\frac{d^2y}{dx^2}\right)\right]$$

**Sol.38. (c)**

$$\frac{dy}{dx} = f(x) + x$$

integrate both side w.r.t. x

$$y = h(x) + \frac{x^2}{2} + c \text{ or } y = g(x) + c$$

both statements are correct.

**Sol.39. (c)**

$$\frac{dx}{dy} + \frac{x}{y} = y^2$$

this equation is L.D.E.

$$\frac{dx}{dy} + Px = Q$$

$$\text{I.F.} = e^{\int \frac{1}{y} dy} = e^{\log y} = y$$

solution of D.E.

$$xy = \int y^2 \cdot y dy$$

$$xy = \frac{y^4}{4} + c$$

$$4xy = y^4 + c$$

**Sol.40. (b)**

$$\frac{ydx - xdy}{y^2} = 0$$

$$d\left(\frac{x}{y}\right) = 0$$

integrate both sides

$$\int d\left(\frac{x}{y}\right) = c$$

$$\frac{x}{y} = c \Rightarrow x = cy \text{ or } y = cx$$

**Sol.41. (a)**

$$\sin\left(\frac{dy}{dx}\right) = a \Rightarrow \frac{dy}{dx} = \sin^{-1} a$$

$$dy = \sin^{-1} a \cdot dx$$

integrate both sides

$$y = x \sin^{-1} a + c$$

**Sol.42. (d)**

maximum power of dy/dx is 5

**Sol.43. (c)**

$$\frac{dy}{dx} = \sqrt{(1-x^2)(1-y^2)}$$

$$\int \frac{dy}{\sqrt{(1-y^2)}} = \int \sqrt{(1-x^2)} dx$$

$$\sin^{-1} y = \frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x + c$$

**Sol.44. (a)**

$$y^2 = 4ax$$

diff. w.r.t. x

$$2yy' = 4a$$

then diff equation

$$y^2 = (2yy')x$$

order 1 and degree 1

**Sol.45. (b)**

circle touching the Y axis at origin

ie centre on X axis (a,0) and its radius is equal to

$$\text{abscissa } (r = a).$$

$$(x-a)^2 + (y-0)^2 = a^2$$

$$x^2 + a^2 - 2ax + y^2 = a^2$$

$$x^2 - 2ax + y^2 = 0 \quad \dots(i)$$

differentiate w.r.t. x

$$2x - 2a + 2y \frac{dy}{dx} = 0$$

$$x - a + y \frac{dy}{dx} = 0$$

$$a = x + y \frac{dy}{dx}$$

put this value in (i)

$$x^2 - 2\left(x + y \frac{dy}{dx}\right)x + y^2 = 0$$

$$x^2 - y^2 + 2xy \frac{dy}{dx} = 0$$

**Sol.46. (d)**

there is only one arbitrary constant so order is 1 but highest degree of c is 4 after removing 3/2 power by squaring

**Sol.47. (a)**

there is only one constant with single power so order and degree both are 1

**Sol.48. (b)**

$$\frac{dx}{dy} + \int y dx = x^3$$

$$\int y dx = x^3 - \frac{dx}{dy}$$

$$1 + \frac{dy}{dx} \left( \int y dx \right) = x^3 \frac{dy}{dx}$$

diff. w.r.t. x

$$0 + \frac{dy}{dx} (y) + \left( \int y dx \right) \left( \frac{d^2 y}{dx^2} \right) = x^3 \frac{d^2 y}{dx^2} + \frac{dy}{dx} (2x^2)$$

$$\frac{dy}{dx} (y) + \left[ x^3 - \frac{dx}{dy} \right] \left( \frac{d^2 y}{dx^2} \right) = x^3 \frac{d^2 y}{dx^2} + \frac{dy}{dx} (2x^2)$$

order 2 and degree 1

**Sol.49. (c)**

y = mx + c (eq. of straight line)

$\frac{dy}{dx} = m$  and mx - y = c is at unit distance from

origin

$$\frac{|m(0) - (0) + c|}{\sqrt{m^2 + 1}} = 1$$

$$c = \sqrt{m^2 + 1}$$

now

$$y = mx + \sqrt{m^2 + 1}$$

$$y - mx = \sqrt{m^2 + 1}$$

$$(y - mx)^2 = m^2 + 1$$

$$\left( y - \frac{dy}{dx} x \right)^2 = \left( \frac{dy}{dx} \right)^2 + 1$$

**Sol.50. (a)**

$$x dy - y dx = y^2 dy$$

$$\frac{y dx - x dy}{y^2} = -dy$$

$$d\left(\frac{x}{y}\right) = -dy$$

$$\int d\left(\frac{x}{y}\right) = -\int dy$$

$$\frac{x}{y} = -y + c$$

$$\text{at } x = 1, y = 1$$

$$1 = -1 + c$$

$$c = 2$$

$$\frac{x}{y} = -y + 2$$

$$\text{at } x = -3$$

$$\frac{-3}{y} = -y + 2$$

by solving it we get -1 and 3

but y > 0 is given in question

so y = 3 is the only answer.

**Sol.51. (b)**

$$\frac{dy}{dx} = \frac{2y}{x}$$

$$\frac{dy}{y} = \frac{2dx}{x}$$

Integrate both sides

$$\int \frac{dy}{y} = \int \frac{2dx}{x}$$

$$\text{Log } y = 2 \log x + \log c$$

$$\text{Log } y = \log x^2 + \log c$$

$$\log y = \log x^2 c$$

$$y = x^2 c$$

$$\text{at point } (1, 1)$$

$$1 = (1)^2 c$$

$$c = 1$$

equation of parabola is y = x<sup>2</sup>

**Sol.52. (a)**

$$\ln\left(\frac{dy}{dx}\right) = a$$

$$\left(\frac{dy}{dx}\right) = e^a$$

$$\int dy = e^a \int dx$$

$$y = xe^{ea} + c$$

**Sol.53. (d)**

$$\text{first put } \left(\frac{dx}{dy}\right) = \frac{1}{\left(\frac{dy}{dx}\right)} \text{ then check degree}$$

**Sol.54. (a)**

$$x dy = y (dx + y dy)$$

$$x dy - y dx = y^2 dy$$

$$y dx - x dy = -y^2 dy$$

$$\frac{y dx - x dy}{y^2} = -dy$$

$$d\left(\frac{x}{y}\right) = -dy$$

integrate both sides

$$\left(\frac{x}{y}\right) = -y + c \quad \dots(i)$$

$$\text{put } x = y = 1$$

$$\text{then } c = 2$$

$$\text{now put } c = 2 \text{ and } x = -3 \text{ in (ii)}$$

$$\text{then } y = 3, -1$$

3 is given in option.

**Sol.55. (a)**

$$y^2 - 2ay + x^2 = a^2$$

diff. w.r.t. x

$$2yy' - 2ay' + 2x = 0$$

$$a = \frac{yy' + x}{y'}$$

$$y^2 - 2\left(\frac{yy' + x}{y'}\right)y + x^2 = \left(\frac{yy' + x}{y'}\right)^2$$

by solving it

$$(x^2 - 2y^2)p^2 - 4pxy - x^2 = 0$$

here p = y'

**Sol.56. (c)**

$$y dx - x dy = 2y^2 dy$$

$$\frac{y dx - x dy}{y^2} = 2dy$$

$$d\left(\frac{x}{y}\right) = 2dy$$

integrate both sides

$$\left(\frac{x}{y}\right) = 2y + c$$

$$x = 2y^2 + cy$$

**Sol.57. (d)**

$$y = A[\sin(x + c) + \cos(x + c)]$$

$$y' = A[\cos(x + c) - \sin(x + c)]$$

$$y'' = A[-\sin(x + c) - \cos(x + c)]$$

$$y'' = -y$$

$$y'' + y = 0$$

**Sol.58. (b)**

Order and degree both are 2

**Sol.59. (d)**

$$\frac{dy}{dx} = \frac{y\phi'(x) - y^2}{\phi(x)}$$

$$\frac{dy}{dx} = \frac{y\phi'(x)}{\phi(x)} - \frac{y^2}{\phi(x)}$$

$$\frac{1}{y^2} \frac{dy}{dx} = \frac{\phi'(x)}{y\phi(x)} - \frac{y^2}{\phi(x)}$$

$$\frac{1}{y^2} \frac{dy}{dx} - \frac{\phi'(x)}{y\phi(x)} = -\frac{1}{\phi(x)}$$

$$\text{put } \frac{-1}{y} = t$$

$$\text{then } \frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + t \cdot \frac{\phi'(x)}{\phi(x)} = -\frac{1}{\phi(x)}$$

$$\text{I.F.} = e^{\int \frac{\phi'(x)}{\phi(x)} dx} = \phi(x)$$

solution of D.E.

$$t \cdot \phi(x) = \int \frac{-1}{\phi(x)} \phi(x) dx$$

$$= -\frac{1}{y} \phi(x) = -x - c$$

$$y = \frac{\phi(x)}{x} + c$$

**Sol.60. (b)**

$$(by + k) dy = (ax + h) dx$$

$$\frac{by^2}{2} + ky = \frac{ax^2}{2} + hx + c$$

$$= \frac{ax^2}{2} - \frac{by^2}{2} - ky + hx + c$$

this equation will represent a circle when a = -b

**Sol.61. (c)**

Order of given D.E. is 3 and degree 2

**Sol.62. (a)**

$$(1 + 2x) dy = (1 - 2y) dx$$

$$\int \frac{1}{1 - 2y} dy = \int \frac{1}{1 + 2x} dx$$

$$-\frac{1}{2} \log(1 - 2y) = \frac{1}{2} \log(1 + 2x) + \frac{1}{2} \log c$$

$$\log(1 + 2x) + \log(1 - 2y) = \log c$$

$$(1 + 2x)(1 - 2y) = c$$

$$2x - 2y - 4xy = c$$

$$x - y - 2xy = c$$

**Sol.63. (a)**

there is only one constant so order is 1

but it is in quadratic form so degree is 2

**Sol.64. (a)**

Given equations are of second order differential equation

if double derivative is negative then solution of these equations will be in the form of sin and cos which are periodic.

**Sol.65. (b)**

$$x dy - y dx = 0$$

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\log y = \log x + \log c$$

$$\log y = \log xc$$

$$y = xc$$

**Sol.66. (c)**

$$\frac{dx}{dy} = \frac{x + y + 1}{x + y - 1}$$

$$\text{put } x + y = t$$

$$\frac{dx}{dy} - 1 = \frac{dt}{dy}$$

$$1 + \frac{dt}{dy} = \frac{t+1}{t-1}$$

$$\frac{dt}{dy} = \frac{t+1}{t-1} - 1$$

$$\frac{dt}{dy} = \frac{2t}{t-1}$$

$$\int \frac{t-1}{t} dt = 2 \int \frac{dy}{y}$$

$$= t - \log t = 2y + c$$

$$= (x+y) - \log(x+y) = 2y + c$$

$$= y - x + \log(x+y) = C$$

**Sol.67. (c)**

$$y = \sin(\ell nx)$$

$$\frac{dy}{dx} = \frac{\cos(\ell nx)}{x}$$

$$x \frac{dy}{dx} = \cos(\ell nx)$$

$$x \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right) = \frac{-\sin(\ell nx)}{x}$$

$$x \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right) = -\frac{y}{x}$$

$$x^2 \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx}\right) + y = 0$$

**Sol.68. (d)**

$$\log\left(\frac{dy}{dx}\right) = ax + by$$

$$\left(\frac{dy}{dx}\right) = e^{ax+by}$$

$$\frac{dy}{dy} = e^{ax} dx$$

integrate both sides

$$-\frac{1}{b} \frac{1}{e^{by}} = \frac{1}{a} e^{ax}$$

$$= \frac{1}{a} e^{ax} + \frac{1}{b} e^{-by} = c$$

**Sol.69. (d)**

there are four arbitrary constant so order is 4.

**Sol.70. (b)**

$$\int dy = \int \left(-x^2 - \frac{1}{x^3}\right) dx$$

$$y = \left(-\frac{x^3}{3} + \frac{1}{2x^2}\right) + c \quad \dots(i)$$

this passes through  $(-1, -2)$

$$-2 = \left(-\frac{1}{3} + \frac{1}{2}\right) + c \Rightarrow c = -\frac{17}{6}$$

put this value in (i)

$$6x^2y + 17x^2 + 2x^5 - 3 = 0$$

**Sol.71. (d)**

$$y = p \cos ax + q \sin ax$$

$$\frac{dy}{dx} = -p \sin ax + q \cos ax$$

$$\frac{d^2y}{dx^2} = -p a^2 \cos ax - q a^2 \sin ax$$

$$\frac{d^2y}{dx^2} = -a^2(p \cos ax + q \sin ax)$$

$$\frac{d^2y}{dx^2} = -a^2 y$$

$$\frac{d^2y}{dx^2} + a^2 y = 0$$

**Sol.72. (a)**

$$\frac{dy}{dx} + \frac{x}{y} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$\int y dy = -\int x dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + c$$

$$x^2 + y^2 = C$$

**Sol.73. (c)**

both are correct.

**Sol.74. (c)**

circle touching the Y axis at origin  
ie centre on X axis (a,0) and its radius is equal to  
abscissa (r = a).

$$(x-a)^2 + (y-0)^2 = a^2$$

$$x^2 + a^2 - 2ax + y^2 = a^2$$

$$x^2 - 2ax + y^2 = 0$$

differentiate w.r.t. x

$$2x - 2a + 2y \frac{dy}{dx} = 0$$

$$x - a + y \frac{dy}{dx} = 0$$

$$a = x + y \frac{dy}{dx}$$

put this value in (i)

$$x^2 - 2\left(x + y \frac{dy}{dx}\right)x + y^2 = 0$$

$$x^2 - y^2 + 2xy \frac{dy}{dx} = 0$$

**Sol.75. (d)**

$$y = a \cos 2x + b \sin 2x$$

$$\frac{dy}{dx} = -2a \sin 2x + 2b \cos 2x$$

$$\frac{d^2y}{dx^2} = -4a \cos 2x - 4b \sin 2x$$

$$\frac{d^2y}{dx^2} = -4(a \cos 2x + b \sin 2x)$$

$$\frac{d^2y}{dx^2} = -4y$$

$$\frac{d^2y}{dx^2} + 4y = 0$$

**Sol.76. (a)**

$$\frac{dy}{dx} = \cos(y-x) + 1$$

put  $(y-x) = t$

$$\frac{dy}{dx} - 1 = \frac{dt}{dx}$$

$$\frac{dt}{dx} + 1 = \cos t + 1$$

$$\frac{dt}{dx} = \cos t$$

$$\int \frac{dt}{\cos t} = \int dx$$

$$\log|\sec t + \tan t| = x + c$$

$$-\log|\sec t - \tan t| = x + c$$

$$\log|\sec t - \tan t| = -x - c$$

$$\sec t - \tan t = e^{-x-c}$$

$$\sec(y-x) - \tan(y-x) = e^{-x} C$$

$$e^x[\sec(y-x) - \tan(y-x)] = C$$

**Sol.77. (c)**

power of highest order differential term is 1

**Sol.78. (b)**

$$\tan y = c(1 - e^x) \quad \dots(i)$$

differentiate both side w.r.t. x

$$\sec^2 y \frac{dy}{dx} = -ce^x$$

put this value of c in (i)

$$\tan y = \frac{\sec^2 y}{-e^x} \frac{dy}{dx} (1 - e^x)$$

cross multiply

$$e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$$

**Sol.79. (d)**

$$y = \frac{1}{2x^2 - c}$$

$$\frac{1}{y} = 2x^2 - c$$

differentiate both sides w.r.t. x

$$-\frac{1}{y^2} \frac{dy}{dx} = 4x$$

$$\frac{dy}{dx} = -4xy^2$$

**Sol.80. (b)**

$$x(dx-dy) + y(dy-dx) = 0$$

$$x dx + y dy = x dy + y dx$$

$$x dx + y dy = d(xy)$$

integrate both sides

$$\frac{x^2}{2} + \frac{y^2}{2} = xy + c$$

$$x^2 + y^2 = 2xy + c$$

**Sol.81. (a)**

$$\ln\left(\frac{dy}{dx}\right) = x$$

$$\frac{dy}{dx} = e^x$$

$$\frac{dx}{dt} = x + 1$$

**Sol.82. (b)**

$$dy = (1 + y^2) dx$$

$$\frac{dy}{1 + y^2} = dx$$

integrate both sides

$$\int \frac{dy}{1 + y^2} = \int dx$$

$$\tan^{-1} y = x + c$$

$$y = \tan(x + c)$$

**Sol.83. (a)**

first differentiate both sides w.r.t. x

then order is 2 and degree is 3

**Sol.84. (a)**

there is only one constant so order of equation is one.

**Sol.85. (a)**

after squaring both sides

degree of D.E. is 1

**Sol.86. (b)**

$$y = ae^x + be^{-x}$$

$$\frac{dy}{dx} = ae^x - be^{-x}$$

$$\frac{d^2y}{dx^2} = ae^x + be^{-x}$$

$$\frac{d^2y}{dx^2} = y$$



$$\frac{d^2y}{dx^2} - y = 0$$

**Sol.87. (c)**

$$\ln\left(\frac{dy}{dx}\right) + y = x$$

$$\ln\left(\frac{dy}{dx}\right) = x - y$$

$$\frac{dy}{dx} = e^{x-y} = \frac{e^x}{e^y}$$

$$e^y dy = e^x dx$$

$$e^y = e^x + c$$

**Sol.88. (c)**

$$\frac{dx}{dt} = x + 1$$

$$\frac{dx}{x+1} = dt$$

$$\log(x+1) = t + c$$

$$\text{at } t = 0, x = 0$$

$$\text{so } c = 0$$

$$\log(x+1) = t$$

$$\text{put } x = 24$$

$$t = \ln 25 = 2 \ln 5$$

**Sol.89. (d)**

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Here is four constant in above equation so order is 4.

**Sol.90. (b)**

$$(x-a)^2 + (y-a)^2 = a^2$$

Maximum power of repeating constants is 2 so degree is 2.

**Sol.91. (b)**

$$y = A - \frac{B}{x}$$

$$y_1 = \frac{B}{x^2}$$

$$y_2 \cdot x^2 \cdot y_1 = B$$

$$x^2 \cdot y_2 + 2x \cdot y_1 = 0$$

$$ny_2 + 2y_1 = 0$$

**Sol.92. (d)**

$$\frac{dy}{dx} = (\log 5)y$$

$$\int \frac{dy}{y} = \int \log 5 \cdot dx$$

$$\text{Log } y = \log 5 \cdot x + \log C$$

$$\text{Log } y = x \log 5 + \log 4$$

$$\text{Log } y = \log 5^x + \log C$$

$$Y = C5^x$$

$$\text{Given } y(0) = \log 5$$

$$\text{Log } 5 = C$$

$$y = \log 5 \cdot 5^x$$

$$= 5 \log 5$$

**Sol.93. (c)**

$$\frac{xdy}{dx} = 2y$$

$$\int \frac{dy}{y} = \int 2 \frac{dx}{x}$$

$$\log y = 2 \log x + \log c$$

$$y = x^2 C$$

$$x^2 = \frac{2}{c} \text{ of equation of parabola.}$$

**Sol.94. (d)**

Degree of given differential equation is 4.

**Sol.95. (a)**

$$\frac{dN}{dt} \propto -N$$

$$\frac{dN}{dt} = -kN$$

$$\int_{N_0}^N \frac{dN}{N} = - \int_0^t k dt$$

$$\log N - \log N_0 = -kt$$

$$\log \frac{N}{N_0} = -kt$$

$$\text{given that } N = \frac{N_0}{2} \text{ at } t = 100 \text{ years}$$

$$\log \frac{1}{2} = -k(100)$$

$$-\log 2 = -100k$$

$$k = \frac{\log 2}{100}$$

**Sol.96. (b)**

Degree does not exist because of  $\cos\left(\frac{dy}{dx}\right)$

Highest derivative is  $\frac{d^2y}{dx^2}$  so order is 2.

**Sol.97. (b)**

Parabola along y axis whose centre is at origin

$$x^2 = 4ay \quad \dots(1)$$

differentiation w.r.t x

$$2x = 4a \frac{dy}{dx} \quad \dots(2)$$

By solving (1) & (2)

$$x^2 = \frac{2x}{\left(\frac{dy}{dx}\right)} y$$

$$x = \frac{dy}{dx} = 2y \Rightarrow x \frac{dy}{dx} - 2y = 0$$

**Sol.98. (c)**

$$dy - dx + \cos x \cdot dy + \cos x \cdot dx = 0$$

$$dy(1 + \cos x) = (1 - \cos x) dx$$

$$dy = \frac{1 - \cos x}{1 + \cos x} dx$$

$$dy = \left( \tan^2 \frac{x}{2} \right) dx$$

$$\int dy = \int \left( \sec^2 \frac{x}{2} - 1 \right) dx$$

$$y = 2 \tan \frac{x}{2} - x + c$$

**Sol.99. (b)**

$$y = (x^x)^x$$

$$y = x^{x^2}$$

$$\log y = x^2 \log x$$

diff. w.r.t x

$$\frac{1}{y} \frac{dy}{dx} = x^2 \cdot \frac{1}{x} + 2x \log x$$

$$\frac{dy}{dx} = y(x + 2x \log x)$$

$$\frac{dy}{dx} = xy(1 + 2 \log x) = 0$$

**Sol.100. (d)**

$$\left( \frac{dy}{dx} \right) \left( \frac{dy}{dx} - x \right) = 0$$

$$\frac{dy}{dx} = 0 \text{ or } \frac{dy}{dx} = x$$

$$y = \text{constant or } \int dy = \int x dx$$

$$y = \text{constant or } y = \frac{x^2}{2} + c$$

for option match put  $C = -1$

**Sol.101. (a)**

Number of arbitrary constants in given equation is only one that is C so order is 1.

**Sol.102. (c)**

Maximum power of only arbitrary constant c is 3/2, after squaring it will be 3 so degree is 3.

**Sol.103. (a)**

$$\frac{dy}{dx} = \frac{2y}{x}$$

$$\frac{dy}{y} = \frac{2dx}{x}$$

Integrate both sides

$$\int \frac{dy}{y} = \int \frac{2dx}{x}$$

$$\text{Log } y = 2 \log x + \log c$$

$$\text{Log } y = \log x^2 + \log c$$

$$\log y = \log x^2 c$$

$$y = x^2 c$$

$$\text{at point } (1, 2)$$

$$2 = (1)^2 c$$

$$c = 2$$

$$\text{equation of parabola is } y = 2x^2$$

$$x^2 = \frac{1}{2} y$$

focal length of this parabola is 1/8 by comparing  $x^2 = 4ay$

equation of directrix of upward parabola is  $y = -a$  so equation is  $y = -1/8$

**Sol.104. (b)**

$$\text{equation of parabola is } y = 2x^2$$

$$x^2 = \frac{1}{2} y$$

focal length of this parabola is 1/8 by comparing

$$x^2 = 4ay$$

$$\text{length of latus rectum} = 4a \text{ i.e. } 1/2$$

**Sol.105. (d)**

order of given D.E. is 3 and degree is 2

**Sol.106. (d)**

$$y^2 = 4a(x-b)$$

differentiate both side w.r.t x

$$2y \frac{dy}{dx} = 4a$$

differentiate both side w.r.t x

$$y \frac{d^2y}{dx^2} + \left( \frac{dy}{dx} \right)^2 = 0$$

**Sol.107. (a)**

$$\frac{dy}{dx} = x$$

$$dy = x dx$$

Integrate both sides

$$\int dy = \int x dx$$

$$y = \frac{x^2}{2} + c$$

$$\text{At } x = 0, y = 0$$

$$\text{Then } c = 0$$

$$y_1(x) \Rightarrow y = \frac{x^2}{2} \quad \dots(i)$$

$$\text{At } x = 0, y = 4$$

$$\text{Then } c = 4$$

$$y_2(x) \Rightarrow y = \frac{x^2}{2} + 4 \quad \dots(ii)$$

By equating both equations

$$\frac{x^2}{2} = \frac{x^2}{2} + 4$$

$0 = 4$  (not possible)

So these two curves never intersect.

**Sol.108. (c)**

$$y = e^x (\cos x + \sin x) \dots\dots\dots(i)$$

differentiate both sides w.r.t.  $x$

$$y' = e^x (\cos x + \sin x) + e^x (-\sin x + \cos x)$$

$$y' = y + e^x (-\sin x + \cos x) \dots\dots\dots(ii)$$

differentiate both side w.r.t.  $x$

$$y'' = y' + e^x (-\sin x + \cos x) + e^x (-\cos x - \sin x) \dots\dots\dots(iii)$$

by using eq (iii) in eq (i)

$$y'' = y' + e^x (-\sin x + \cos x) - y$$

by using eq (ii)

$$y'' = y' - y + y' - y$$

$$y'' - 2y' + 2y = 0$$

**Sol.109. (d)**

$$\left\{ 2 - \left( \frac{dy}{dx} \right) \right\}^{0.6} = \frac{d^2 y}{dx^2}$$

Power 0.6 means  $3/5$

By taking 5 power both sides

$$\left\{ 2 - \left( \frac{dy}{dx} \right) \right\}^3 = \left( \frac{d^2 y}{dx^2} \right)^5$$

By above D.E. order is 2 and degree is 5

**Sol.110. (a)**

$$\frac{dy}{dx} = 2e^x y^3$$

$$\frac{dy}{y^3} = 2e^x dx$$

Integrate both sides

$$\int \frac{dy}{y^3} = \int 2e^x dx$$

$$\frac{-1}{2y^2} = 2e^x + c$$

Given that at  $x=0$ ,  $y = 1/2$

$$-2 = 2 + c$$

$$c = -4$$

$$\frac{-1}{2y^2} = 2e^x - 4$$

$$-1 = 2y^2(2e^x - 4)$$

$$1 = 4y^2(2 - e^x)$$

**Sol.111. (a)**

$$y dx + (x - y^3) dy = 0$$

$$y dx = -(x - y^3) dy$$

$$\frac{dy}{dx} = \frac{-y}{x - y^3}$$

so order 1 and degree

**Sol.112. (d)**

$$y dx + (x - y^3) dy = 0$$

$$y dx + x dy - y^3 dy = 0$$

$$y dx + x dy = y^3 dy$$

$$d(xy) = y^3 dy$$

integrate both sides

$$\int d(xy) = \int y^3 dy$$

$$xy = \frac{y^4}{4} + C$$

$$4xy = y^4 + 4C$$

**Sol.113. (a)**

$$\text{slope} = \frac{dy}{dx} = 4$$

$$dy = 4dx$$

$$\int dy = \int 4dx$$

$$y = 4x + c$$

passing through origin so  $c = 0$

$y = 4x$  (straight line passing through origin and satisfying (1, 4)).

**Sol.114. (a)**

$$\left( \frac{d^2 y}{dx^2} \right)^{\frac{3}{2}} = \left( \frac{dy}{dx} \right)^{\frac{5}{2}}$$

Whole square both side

$$\left( \frac{d^2 y}{dx^2} \right)^3 = \left( \frac{dy}{dx} \right)^5$$

Degree of above D.E. is 3

**Sol.115. (c)**

$$y = xe^{2x}$$

$$\frac{dy}{dx} = e^{2x} + 2xe^{2x}$$

$$= xe^{2x} \left( \frac{1}{x} + 2 \right) = y \left( 2 + \frac{1}{x} \right)$$

Statement 1 is correct

$$y = x \ln x + cx$$

$$\frac{dy}{dx} = \ln x + x \cdot \frac{1}{x} + c$$

$$\frac{dy}{dx} = 1 + \ln x + c = \frac{x + x \ln x + cx}{x}$$

$$= \frac{x + y}{x}$$

Statement 2 is correct

**Sol.116. (b)**

$$(x + y)^2 \frac{dy}{dx} = k^2$$

$$\frac{dy}{dx} = \frac{k^2}{(x + y)^2}$$

$$\text{Let } x + y = t, \text{ then } 1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - 1 = \frac{k^2}{t^2}$$

$$\frac{dt}{dx} = \frac{k^2}{t^2} + 1 = \frac{k^2 + t^2}{t^2}$$

$$\frac{t^2}{k^2 + t^2} \frac{dt}{dx} = 1$$

$$\frac{t^2 + k^2 - k^2}{k^2 + t^2} dt = dx$$

$$1 - \frac{k^2}{k^2 + t^2} dt = dx$$

$$\int 1 - \frac{k^2}{k^2 + t^2} dt = \int dx$$

$$t - k^2 \frac{1}{k} \tan^{-1} \left( \frac{t}{k} \right) = x + c$$

$$x + y - k \tan^{-1} \left( \frac{x + y}{k} \right) = x + c$$

$$y - k \tan^{-1} \left( \frac{x + y}{k} \right) = c$$

$$\frac{y - c}{k} = \tan^{-1} \left( \frac{x + y}{k} \right)$$

$$\tan \left( \frac{y - c}{k} \right) = \left( \frac{x + y}{k} \right)$$

$$x + y = k \tan \left( \frac{y - c}{k} \right)$$